Information Technology Institute

Generative AI

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Who am I

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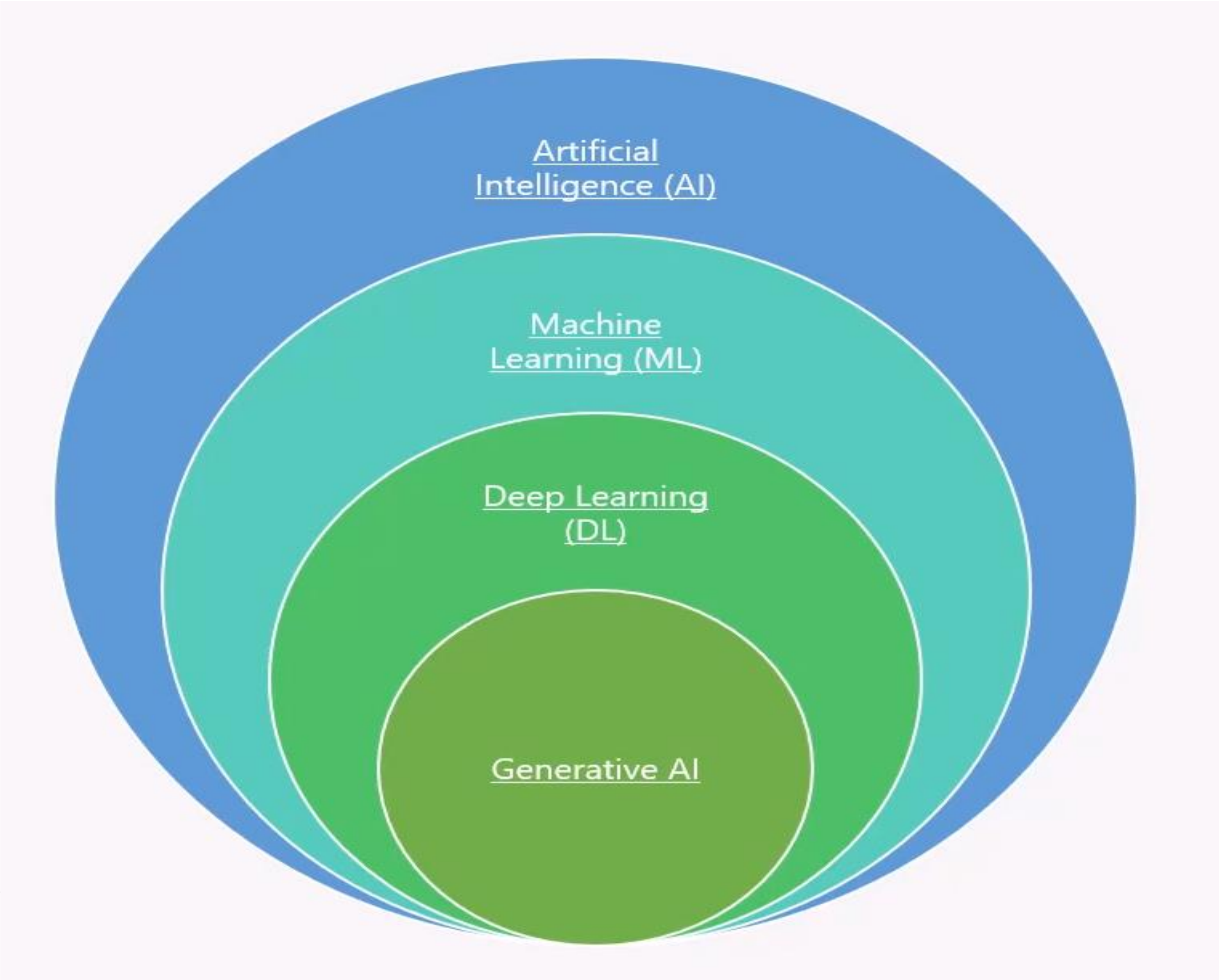
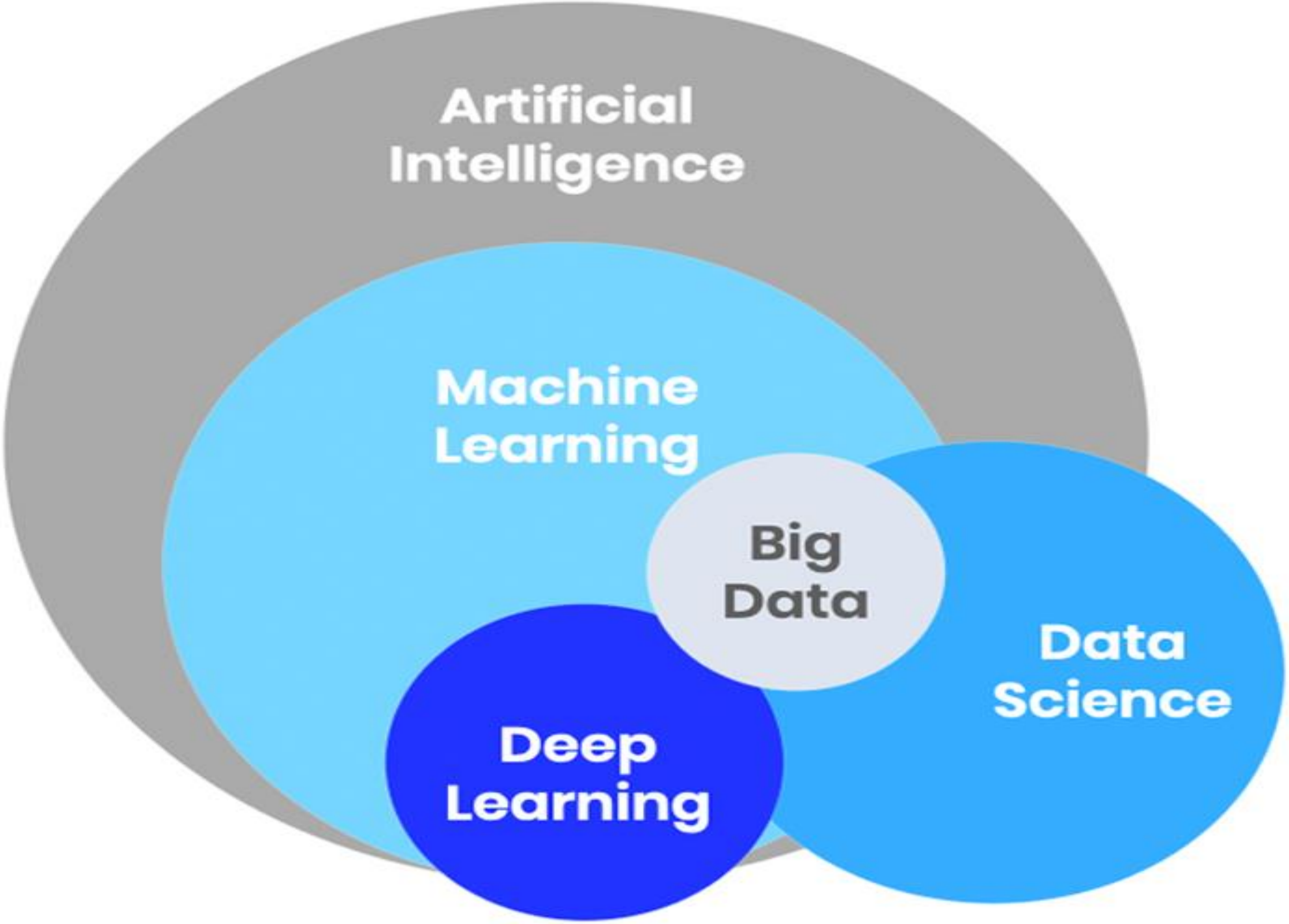
Ministry of Communication & Information Technology

Course Outline

- Intro to Gen AI: evolution of AI to Generative AI.
- Gen AI benefits and practical use cases
- Key Gen AI terminologies and merging technologies:
- Gen AI tools models, tools (ChatGPT, Gemini, Claude, MS Copilot, DeepSeek, Qwen, Grok…) and use cases in different fields.
- Introducing prompt engineering
- Use cases for Gen AI applications and uses (to be tailored for each track)
- AI tools limitations, security and data privacy considerations

Artificial Intelligence and Generative AI?





These terms are closely related areas within the field of artificial intelligence.

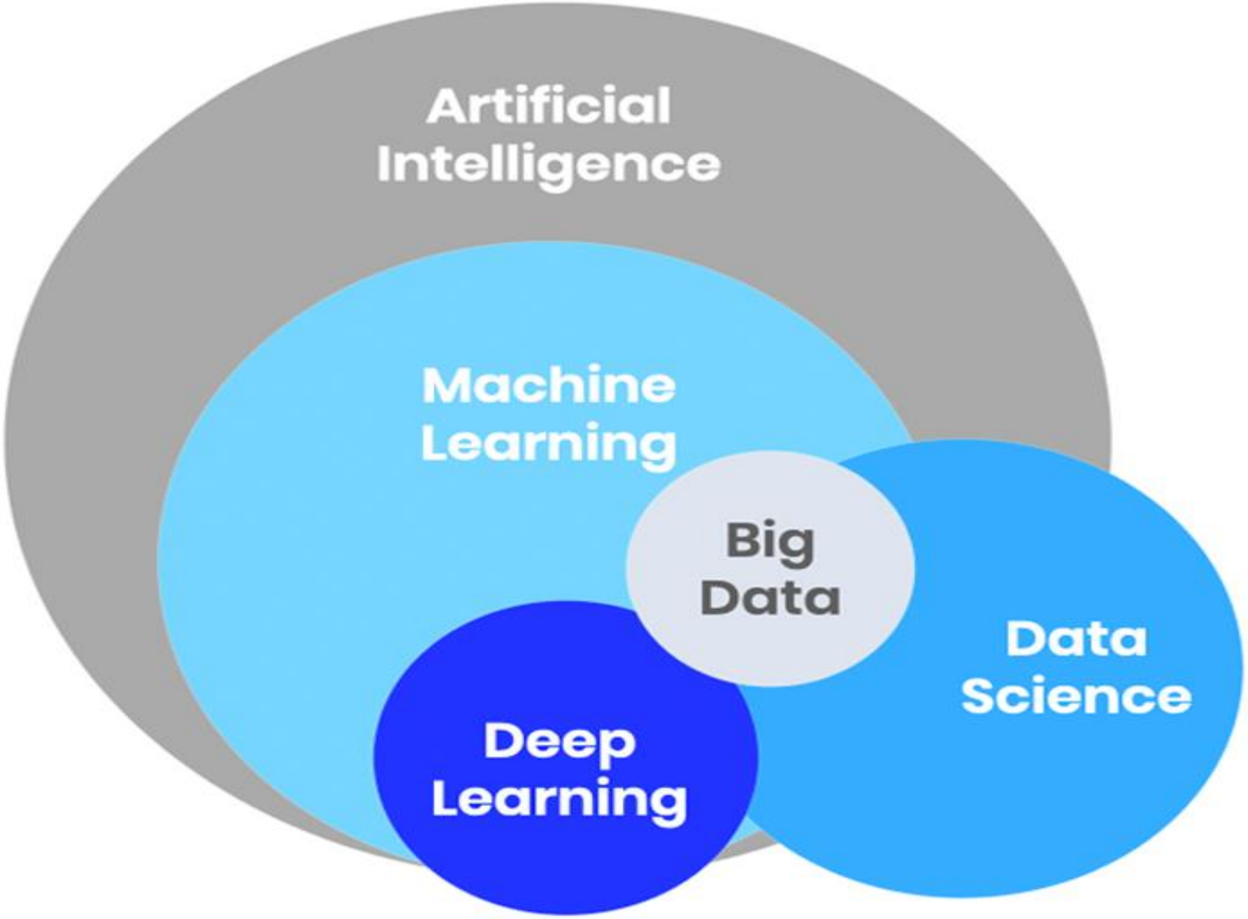
All of these terms: AI, Data Science, Machine Learning, Deep Learning and Big Data are interlinked (مترابطة)

What is artificial intelligence, machine learning, deep learning, Data science, Big Data?

Artificial Intelligence

What is artificial intelligence?

You can think of artificial intelligence as an umbrella term that refers to any system that **mimics human behavior or intelligence**, no matter how simple or complicated that behavior is.

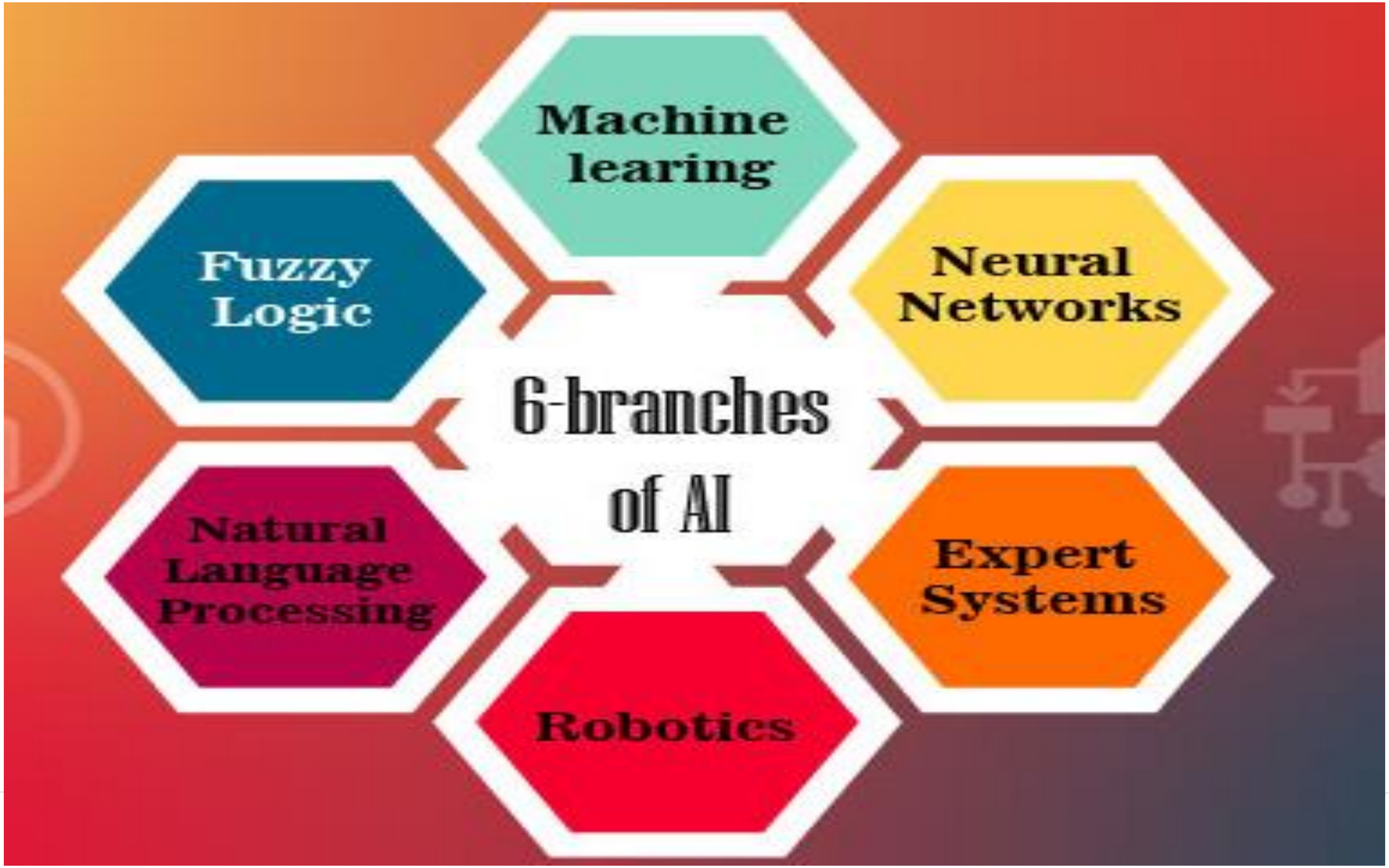


Artificial Intelligence is any system that mimics human behavior or intelligence

Artificial Intelligence (AI) is defined as a branch of computer science that aims to enable computer systems to perform various tasks with intelligence similar to humans.

AI is the science of building machine intelligence that is capable of thinking like humans. It uses Algorithms and methods to build machines that can perform human-like tasks.

Major branches of Artificial Intelligence



Types of Artificial Intelligence

Artificial Intelligence can be divided in various types According to the capabilities:

(بناءاً على قدرة الذكاء الاصطناعي على تنفيذ المهام)

- **Narrow AI, or Weak AI**
- **General AI**
- **Super AI**

Types of Artificial Intelligence

Based on capabilities, AI can be divided into three types that are:

Narrow AI:

is a type of AI which is able to perform a **dedicated task** with intelligence. Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as **weak AI**.

General AI:

The idea behind the general AI to make such a system which could be smarter and **think like a human and perform any task as perfect as a human**.

Super AI:

is a level of Intelligence of Systems at which **machines could exceed human intelligence**, and can perform any task better than human.

Types of Artificial Intelligence

Type-1: Based on Capabilities

At the current stage, AI is known as **Narrow AI or Weak AI**, which can only perform dedicated tasks.

For example, **self-driving cars, Chatbots,** etc.

ChatGPT is considered a form of Narrow AI because it's limited to the single task of text-based chat.

Types of Artificial Intelligence

Based on capabilities, AI can be divided into three types that are:

Narrow AI:

The most **common and currently available AI is Narrow AI** in the world of Artificial Intelligence.

General AI:

Currently, **there is no such system exist (general AI are still under research)** and The worldwide researchers are now focused on developing machines with General AI. And it will take lots of efforts and time to develop such systems.

Super AI:

It is an **outcome of general AI**.

Machine Learning

Narrow Artificial Intelligence (ANI)

Stage One: Machines imitate human behavior, specializing in one area to solve a problem.

i.e. Siri, ChatGPT, Alexa

Machine Intelligence

Artificial General Intelligence (AGI)

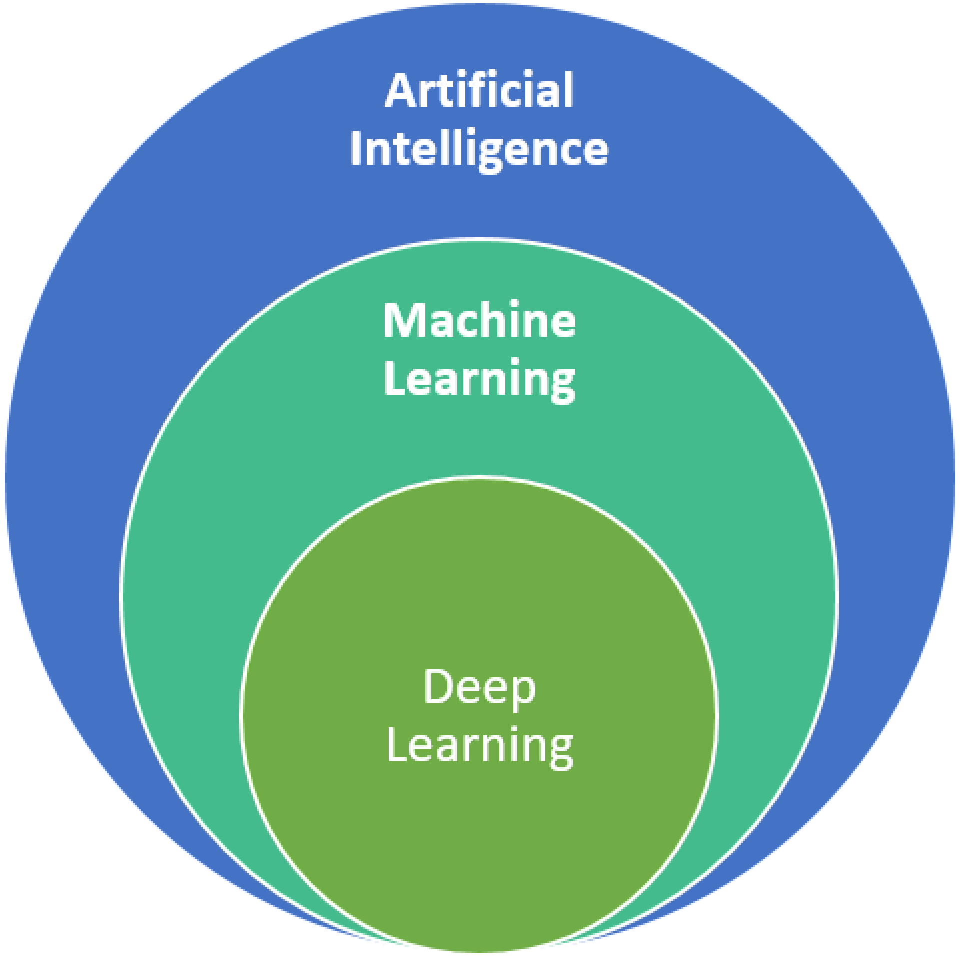
Stage Two: Machines can continuously learn and are as smart as humans.

Machine Consciousness

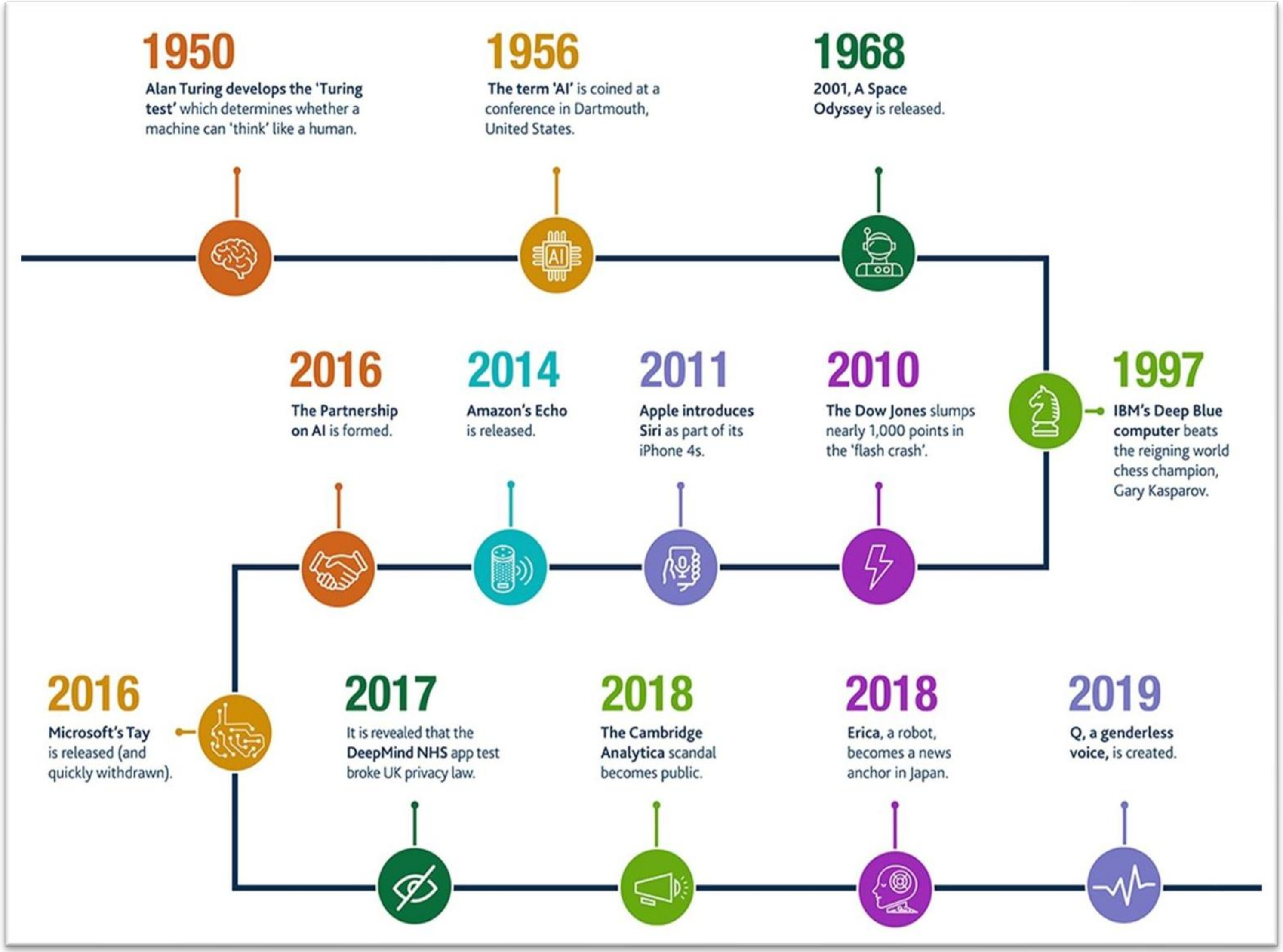
Artificial Super Intelligence (ASI)

Stage Three: Machines that are smarter than humans across the board.

artificial intelligence, machine learning, deep learning



History of AI



🧠 The Story of AI — From Logic to Creativity

1950s – The Idea Begins

Alan Turing asked: "*Can machines think?*"

This question started the journey of Artificial Intelligence.

1960s–1980s – Symbolic AI

AI was built on hand-coded rules and logic.

Good at solving specific problems, but couldn't handle uncertainty or complex data.

1990s–2000s – Machine Learning

Instead of coding all rules, machines started learning from data.

This allowed better accuracy and more flexibility.

2010s – Deep Learning Revolution

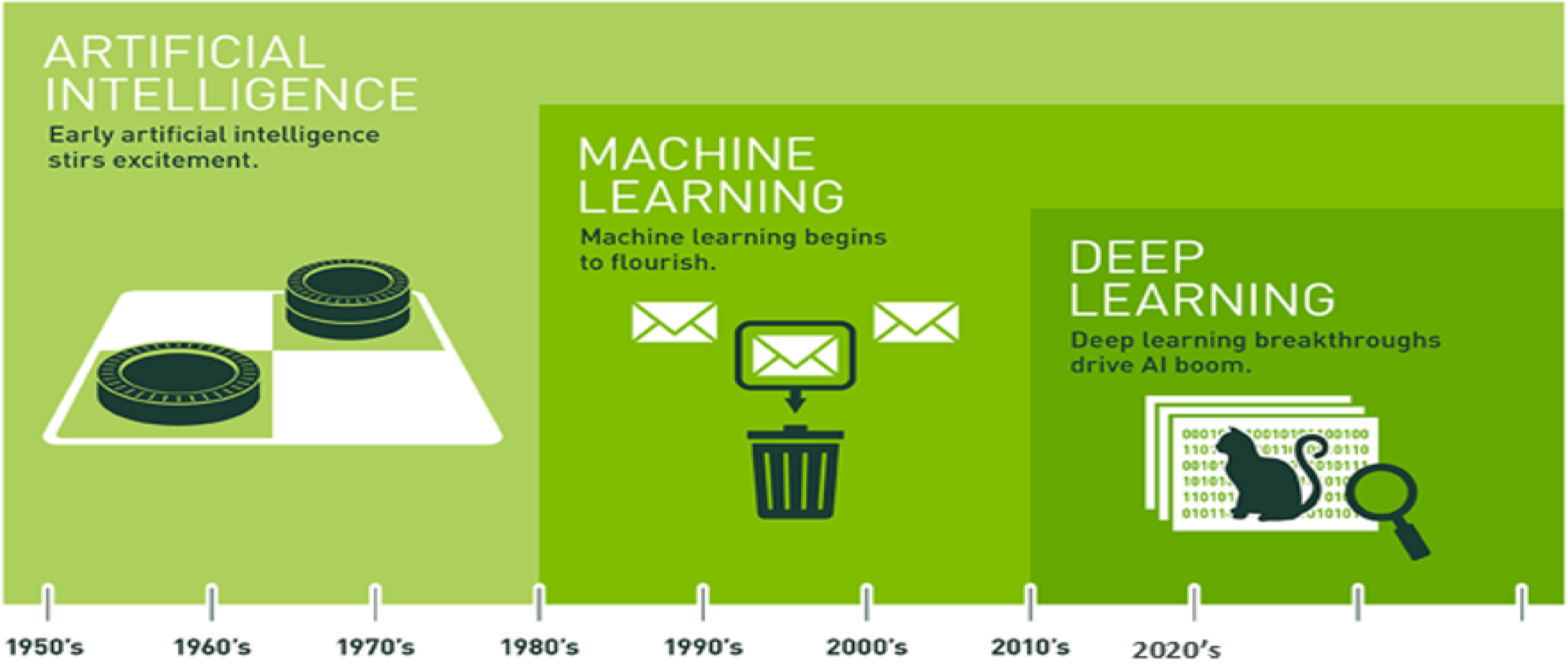
Neural networks became deeper and more powerful with the rise of GPUs.

AI began understanding images, speech, and language at scale.

2020s – Generative AI (GenAI)

Now, AI doesn't just *understand* — it *creates*.

artificial intelligence, machine learning, deep learning



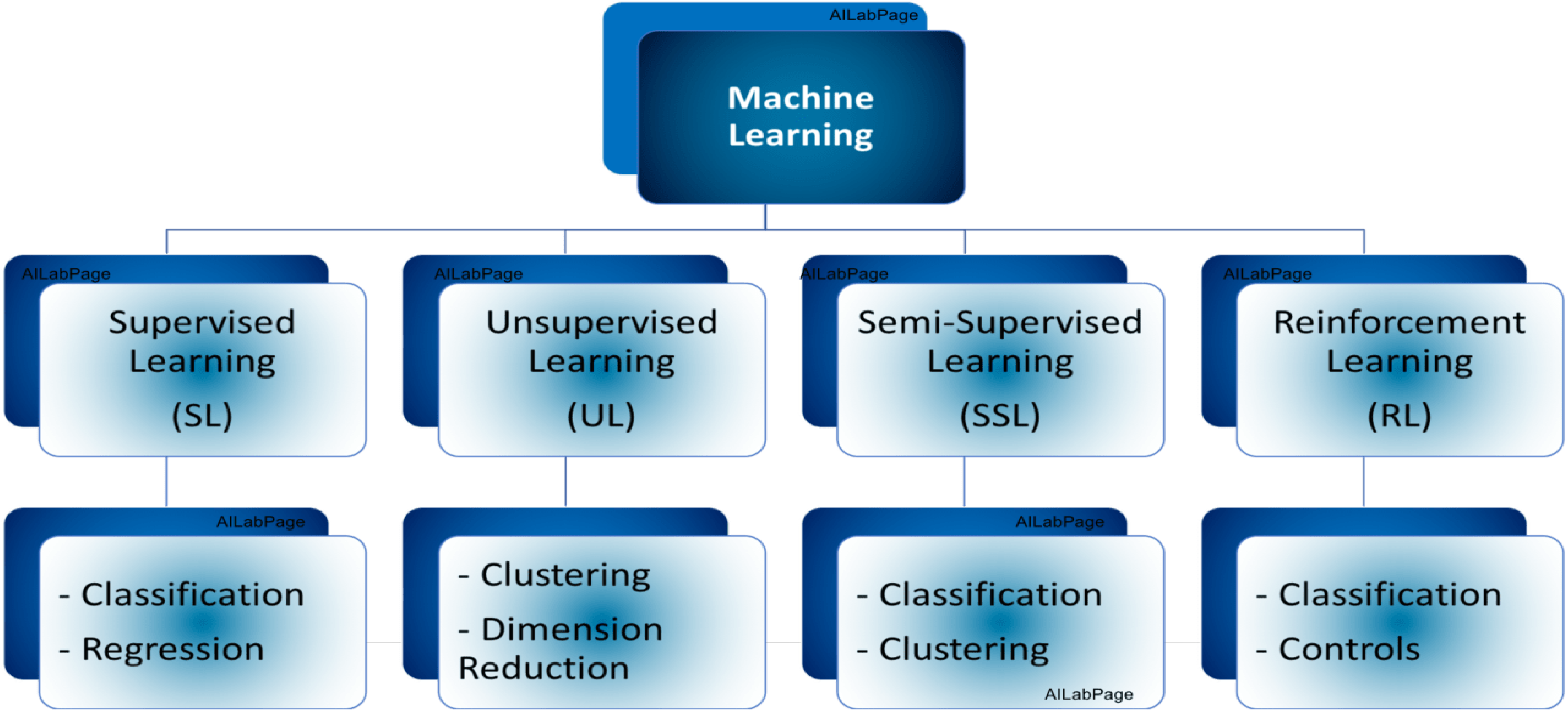
Machine Learning

What is machine learning

machine learning is a subset of artificial intelligence (AI), which focuses on using statistical techniques to build intelligent computer systems to learn from available databases.

It focuses on the use of data and algorithms to mimic the way that humans learn, gradually improving its accuracy.

Types of machine learning



Types of machine learning

Supervised learning: Supervised learning gets labeled inputs and their desired outputs. The goal is to learn a general rule to map inputs to the output.

Unsupervised learning: Machine gets inputs without desired outputs, the goal is to find structure or hidden patterns for useful information in inputs. Extremely useful when we have a very large amount of unlabeled data.

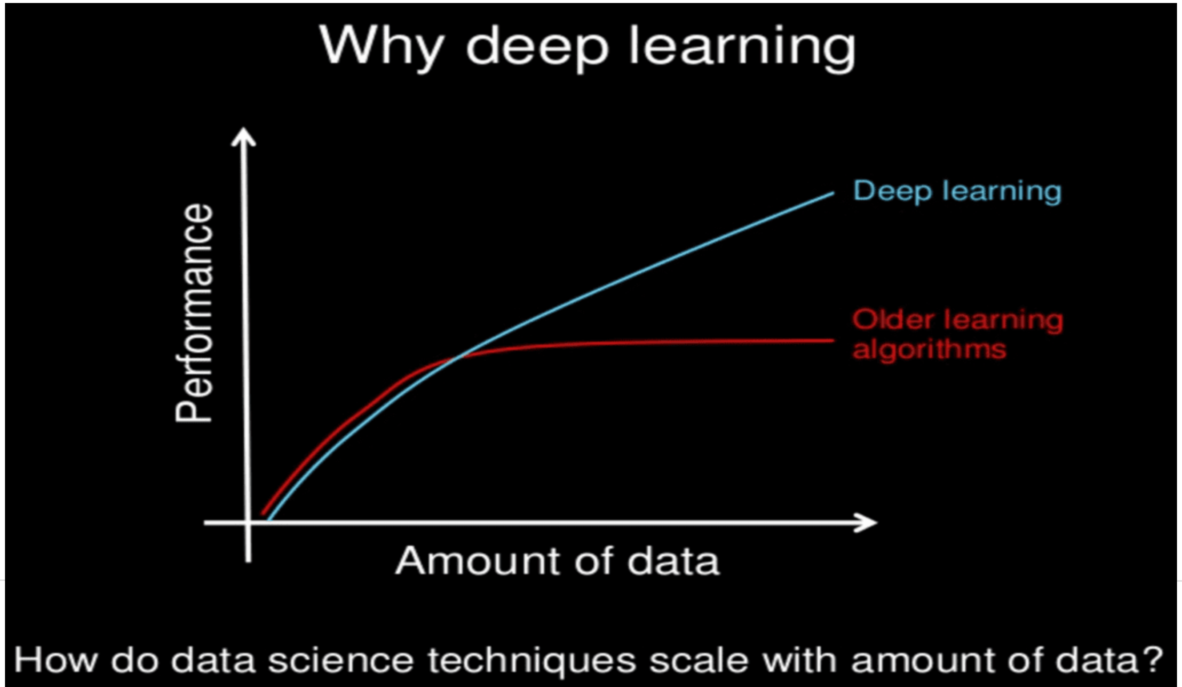
Semi-supervised Learning: This type is the best model in the absence of labels for some data. So if data is a mix of labels and non-labels then this can be the answer. Typically a small amount of labeled data with a large amount of unlabeled data is used here.

Reinforcement learning: In this algorithm interacts with a dynamic environment, and it must perform a certain goal without a guide or teacher.

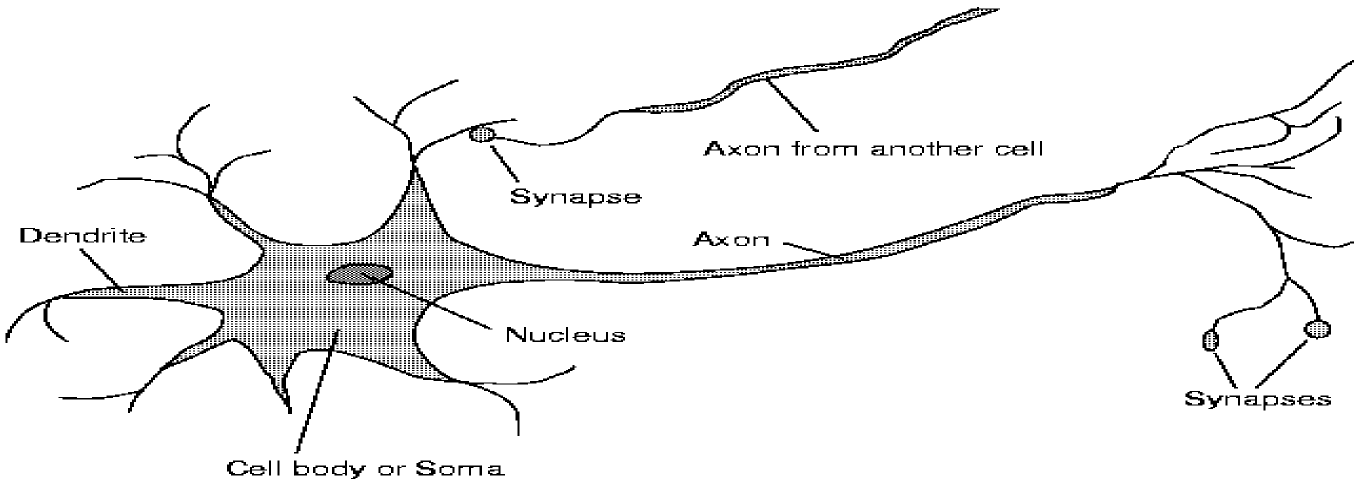
Deep Learning

What is deep learning

- **deep learning** is a subset of machine learning, which refers to a technique in which a system uses neural networks that **mimic the activities of a human brain** to make intelligent decisions.



The brain is composed of **neurons**



We are born with about **100 to 200 billion neurons**

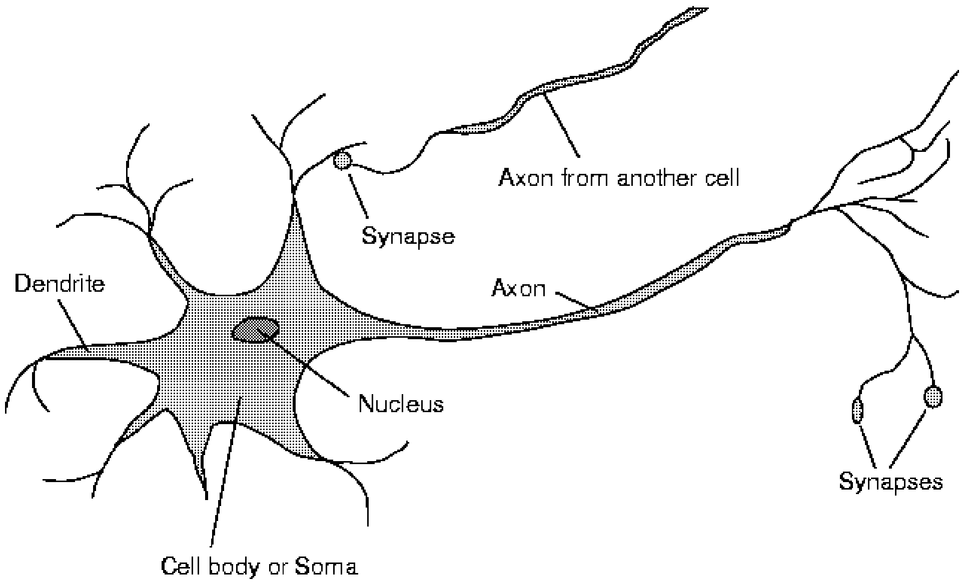
A neuron may connect to as many as **100,000** other neurons

Many neurons die as we progress through life

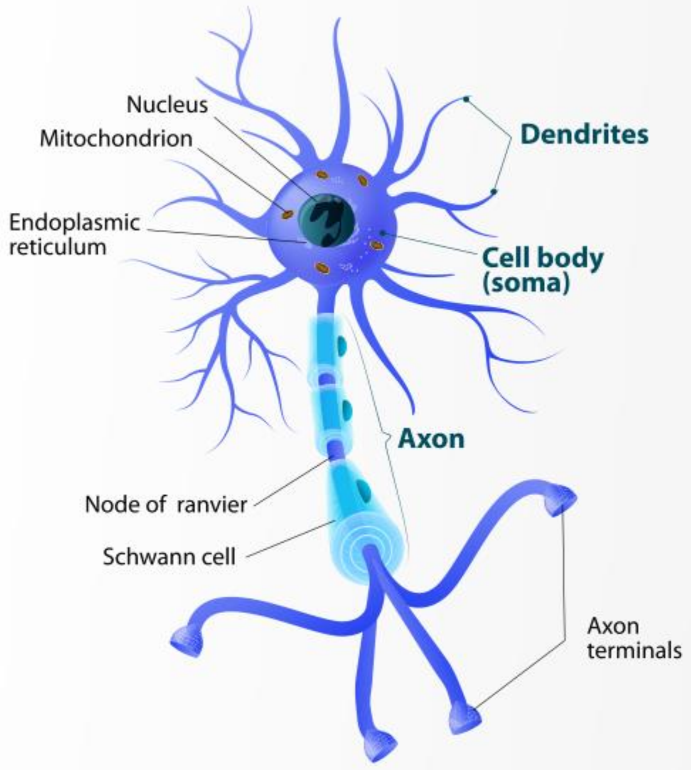
The First Artificial Neural Networks

Consisted of:

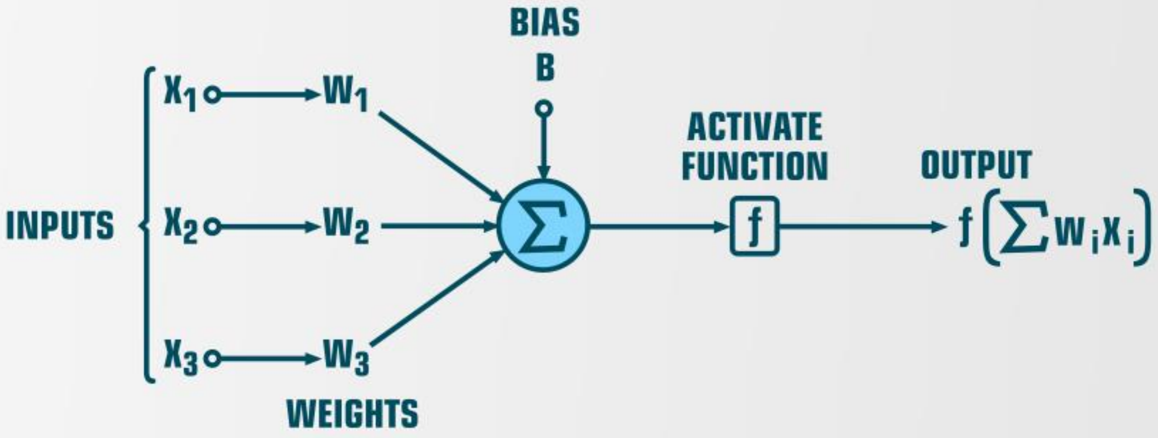
- A set of inputs - (dendrites)
- A set of resistances/weights – (synapses)
- A processing element - (neuron)
- A single output - (axon)



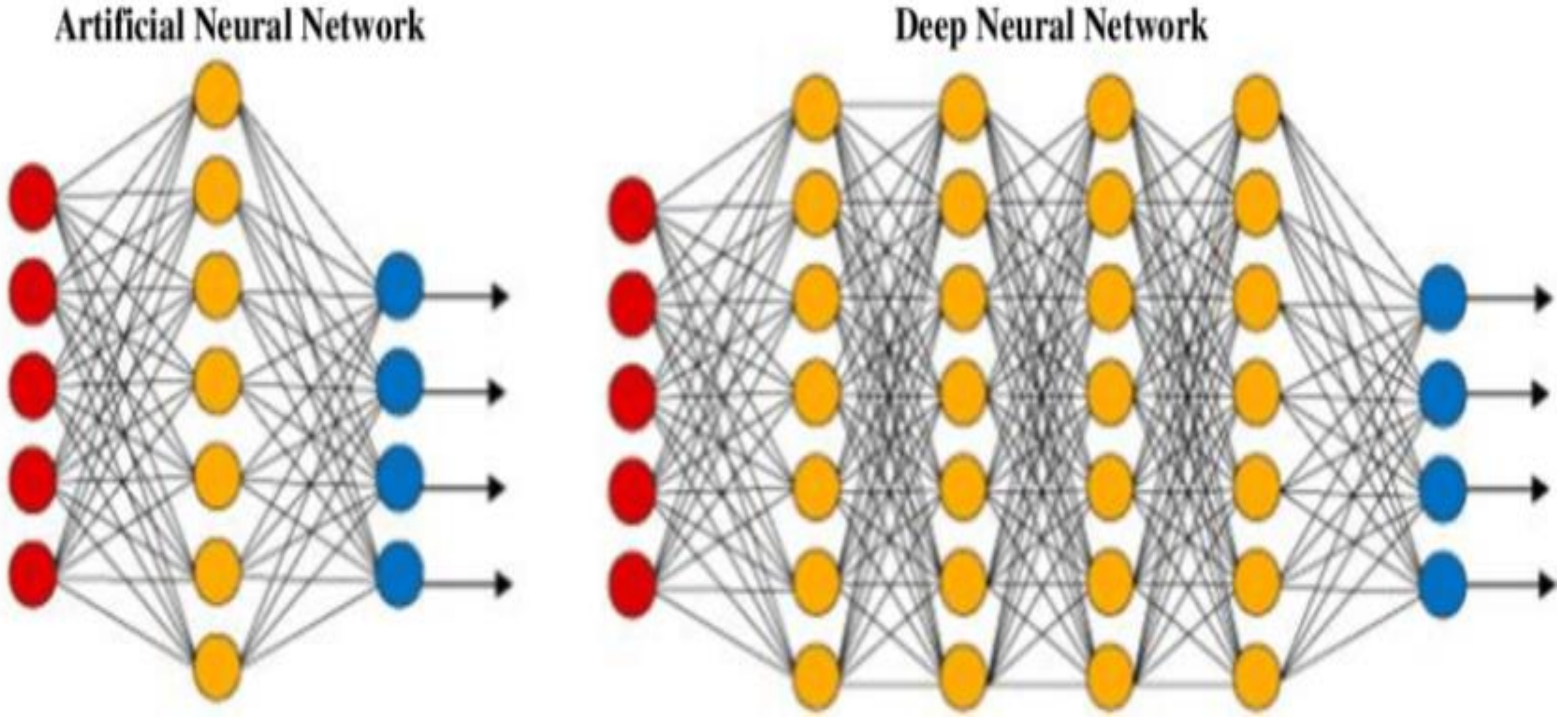
Structure of Typical Neuron



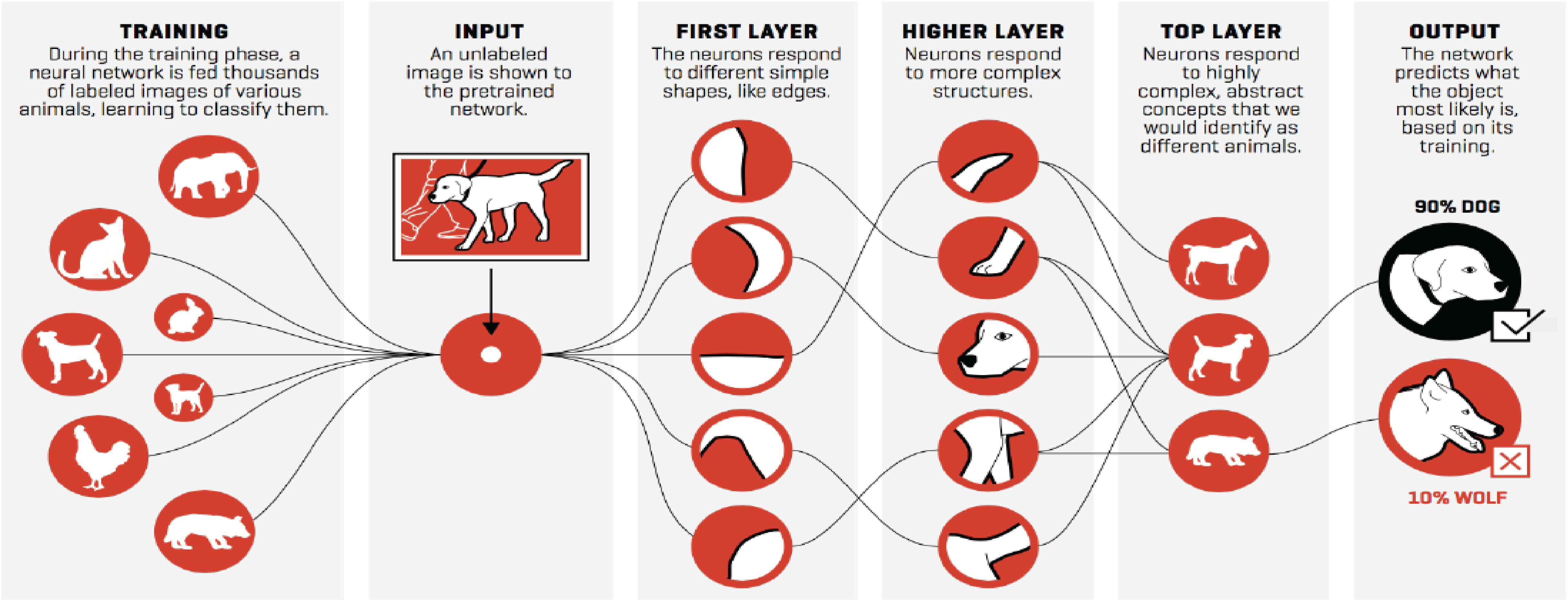
Structure of Artificial Neuron



Neutral Network



Deep Learning Example



Generative AI

What is Generative AI?

Applications of GenAI in Different Fields

Introduction to Prompt Engineering

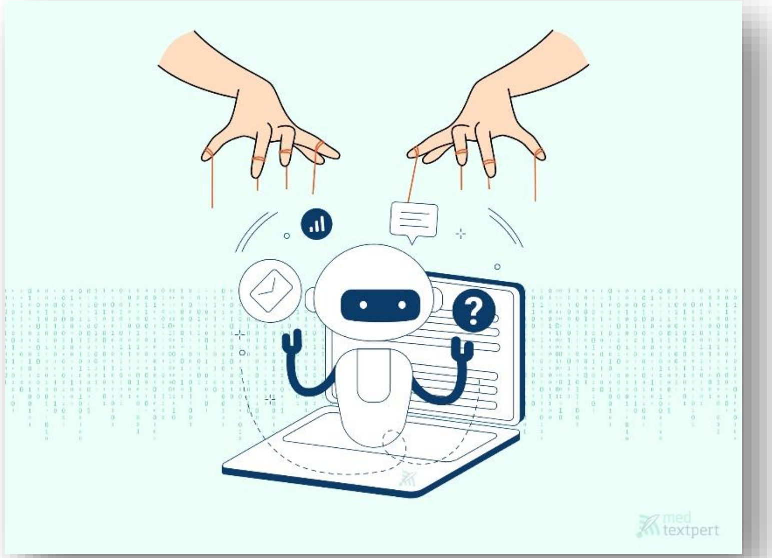
Examples of GenAI Tools

Real Projects

Risks and Ethics

Generative AI

- Generative AI creates new content like text, images, and music from data.
- It learns patterns from huge datasets to produce human-like output.
- With the right prompt, Generative AI can write, draw, or even code.

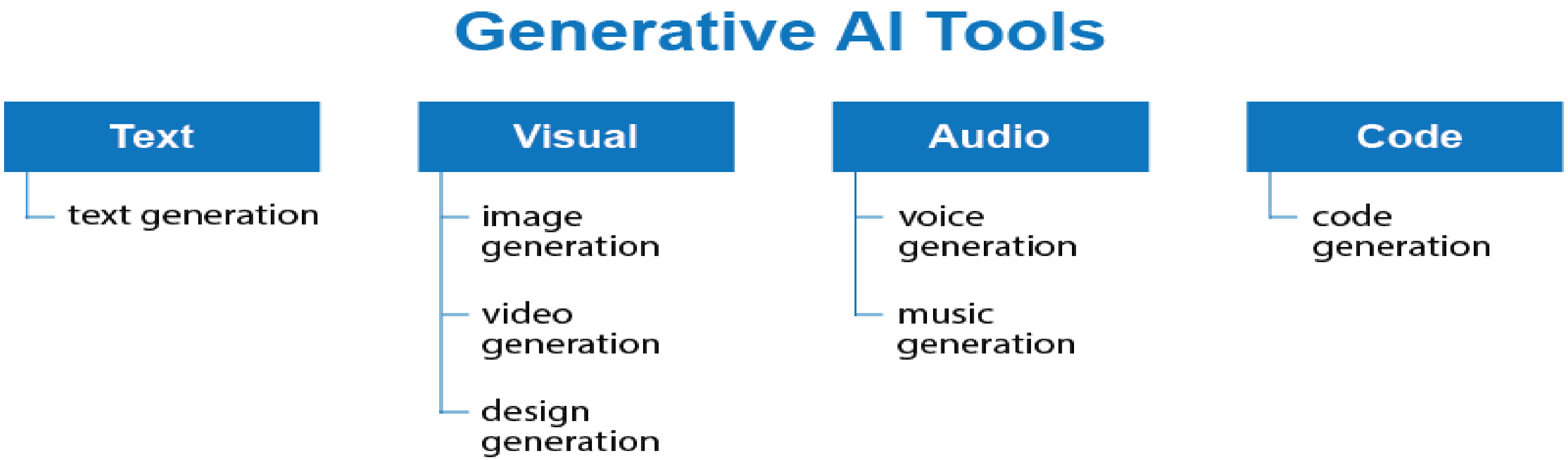


Why do we need to know about Gen AI?

What's Generative AI?

Generative AI: is a type / subset of artificial intelligence system that can produce various types of content, including text, imagery, audio and synthetic data.

Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.



Generative AI

Myth	Reality
"Generative AI is always correct."	AI can confidently generate false or misleading content.
"GenAI can think or feel."	AI has no emotions, consciousness, or self-awareness.
"It creates completely original content."	It recombines existing data in new ways—not always truly creative.
"It will replace all human jobs."	GenAI supports humans—it doesn't replace creativity, empathy, or judgment.
"AI-generated content is always safe to use."	It may include bias, inappropriate, or copyrighted information.
"Only tech experts can use it."	Today's tools are user-friendly for writers, students, designers, and more.

Generative AI

AI is evolving FAST, and many people fear that it will replace human workers.

But here's the truth they won't tell you:

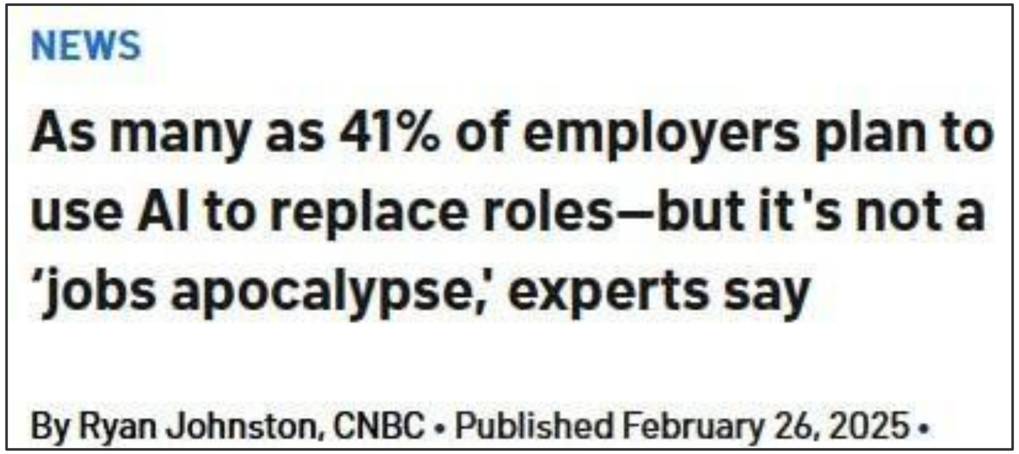
AI isn't replacing JOBS—it's replacing TASKS.

And those who adapt and upskill will be the ones who THRIVE in the AI era. 🚀

https://www.instagram.com/maa.academy.blr/reel/DHwBO8RPQtd/

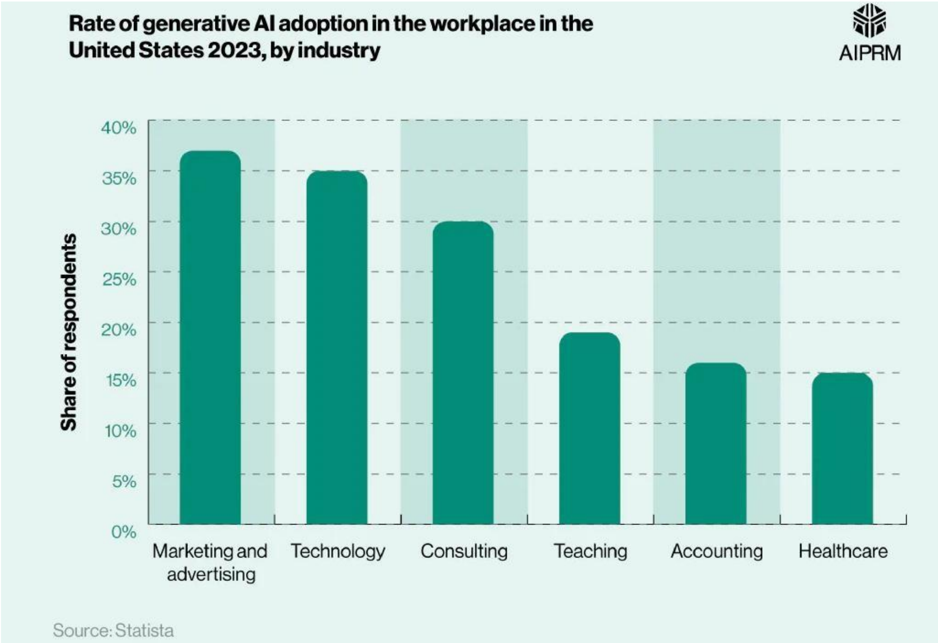
THRIVE

Will AI Replace Jobs / employees?


NEWS

As many as 41% of employers plan to use AI to replace roles—but it's not a 'jobs apocalypse,' experts say

By Ryan Johnston, CNBC • Published February 26, 2025 •

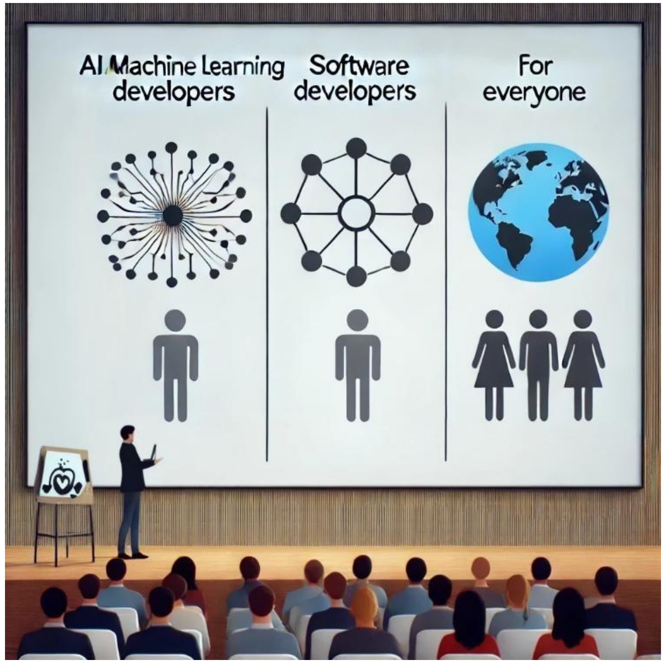


Who should learn about Gen AI?

- **AI / Machine Learning Developers**:
  - Design and refine Generative AI models to create more intelligent and efficient AI-driven solutions.

- **Software Developers**:
  - Utilize AI tools to enhance productivity in development tasks.
  - And use Generative AI to build AI-powered applications

- **Everyone**:
  - In today's AI-centric world, the individuals across all professions have to understand the basics of Generative AI and merge AI tools into their workflows to increase productivity.



Generative AI Tools (Examples)

🧠 Text Generation

- ChatGPT writing essays, summaries, or poetry
- Jasper creating marketing copy

🎨 Image Generation

- DALL·E generating artwork from text prompts
- Midjourney creating stylized illustrations or concept art

🎵 Music Creation

- Aiva composing classical-style music
- Suno.ai generating songs from a theme or lyrics

🎬 Video & Animation

- Runway ML turning text into short video clips
- Pika Labs generating animated scenes from prompts

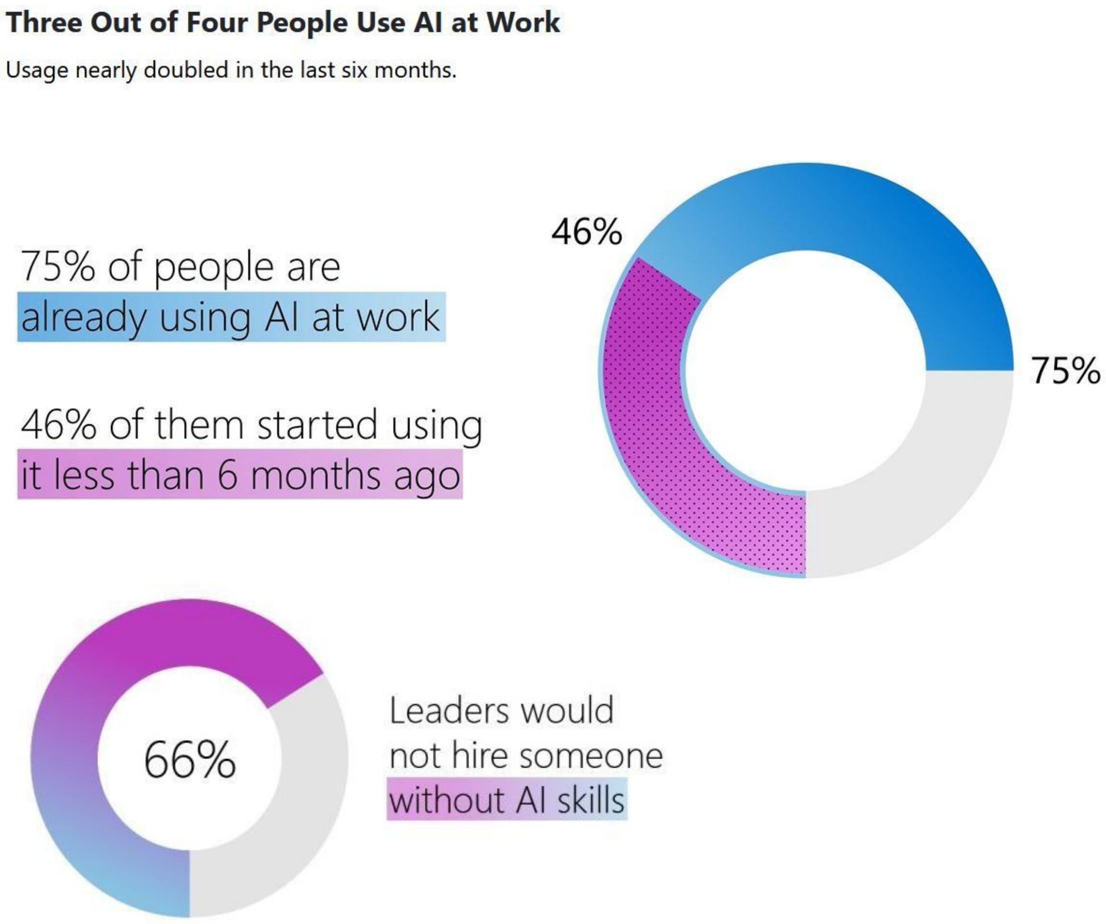
🧑‍💻 Code Generation

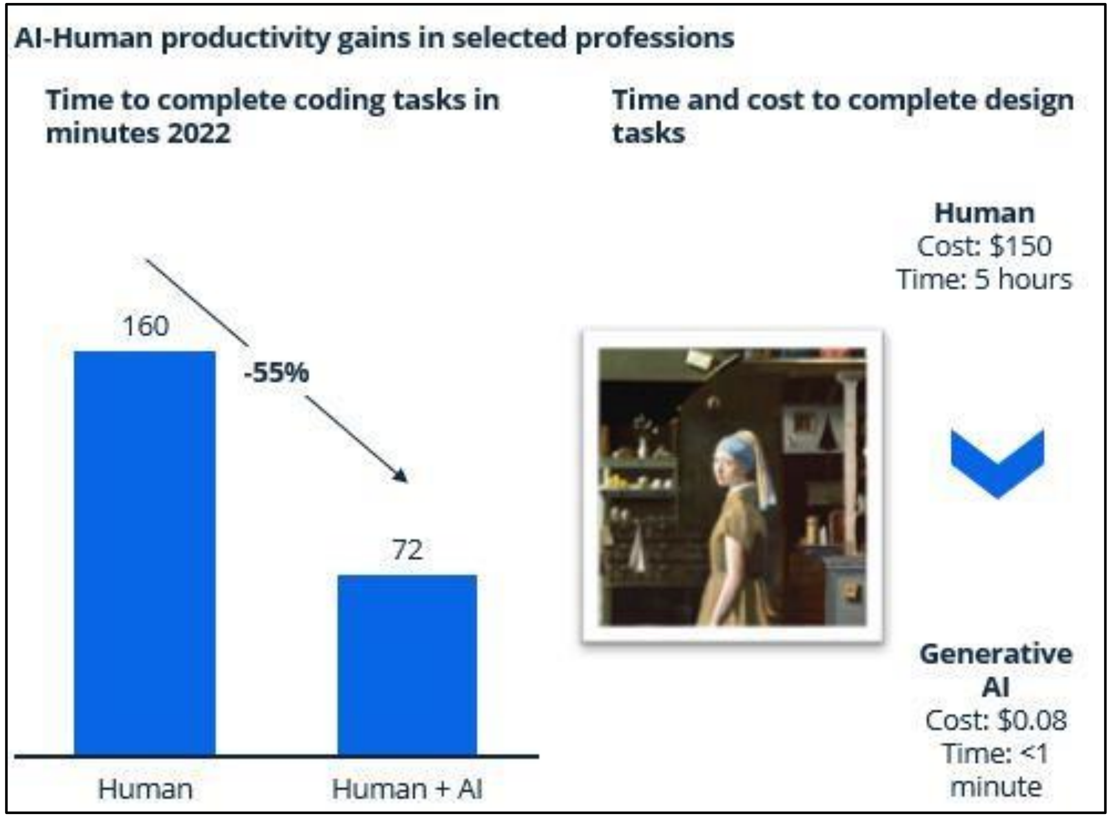
- GitHub Copilot suggesting code snippets inside editors
- Replit Ghostwriter building small applications or debugging

🗣️ Speech & Voice

- ElevenLabs cloning voices or generating speech from text
- Descript Overdub editing audio with AI-generated voice

Gen-AI Tools in Work Statistics





2024 Work Trend Index Annual Report from Microsoft and LinkedIn:
AI In Work

[AI Improves Employee Productivity by 66%]

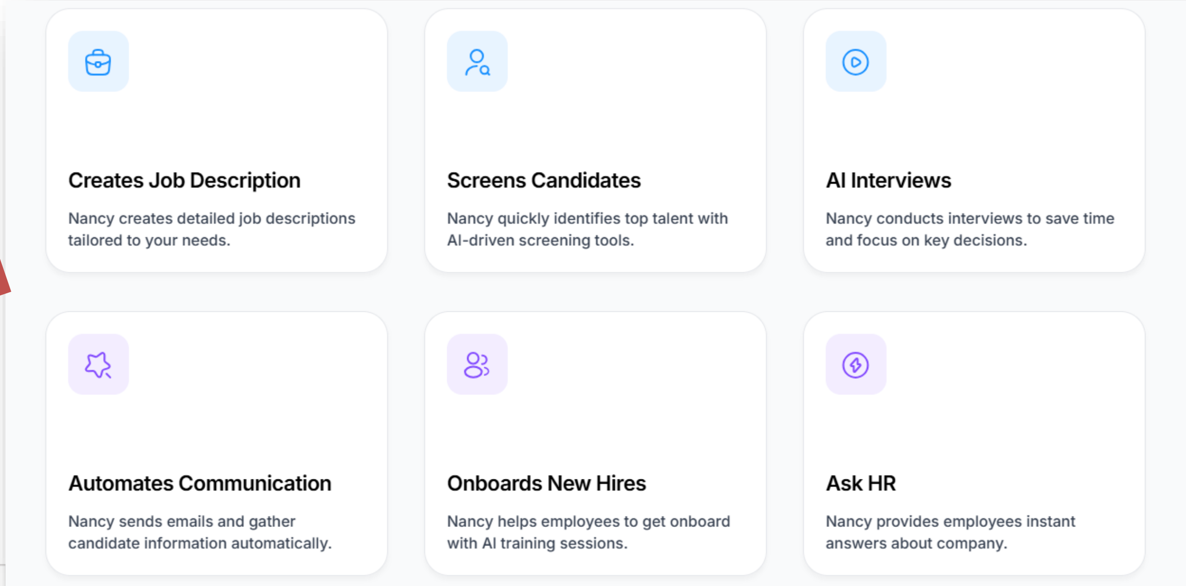
Sample GenAI Freelancing jobs

- **Upwork**
  - AI Image Generation and Deepfake Model Creation
  - AI developer (rag + llm) + fullstack
  - Next.js Developer Needed for SaaS Application Integration with AI Features
  - Developers for 5 AI agents
  - AI Engineer for Medical RAG-based Chatbot Development
  - GenAI engineer to build GenAI-powered modules of an astrology app
  - CReate AI Agent using Open AI or Google AI API or CrewAI
  - Full-Stack Developer with AI Experience (Replit Preferred)
  - Developer Needed to Build AI Agents for Data Analysis and Digital Advertising
  - Web App Development with AI for Meta Ads Management
  - ... and more

- **Khamsat and Mostaql**
  - Khamsat Gen AI Services
  - Mostaql Gen AI Projects

Gen-AI based startups and real products

- **Use cases for real products using Gen AI:**
  - Loop-x (AI Agents for different roles): https://loop-x.co
  - Gemelo ai (Twin AI creator): https://gemelo.ai
  - Apriora AI (Recruitment and Interviews): https://www.apriora.ai
  - Velents AI (Recruitment and Interviews): : https://www.velents.com
  - Nancy AI (Recruitment and Interviews): : https://nancy-ai.com
  - Mariana AI (Medical): https://marianaai.com



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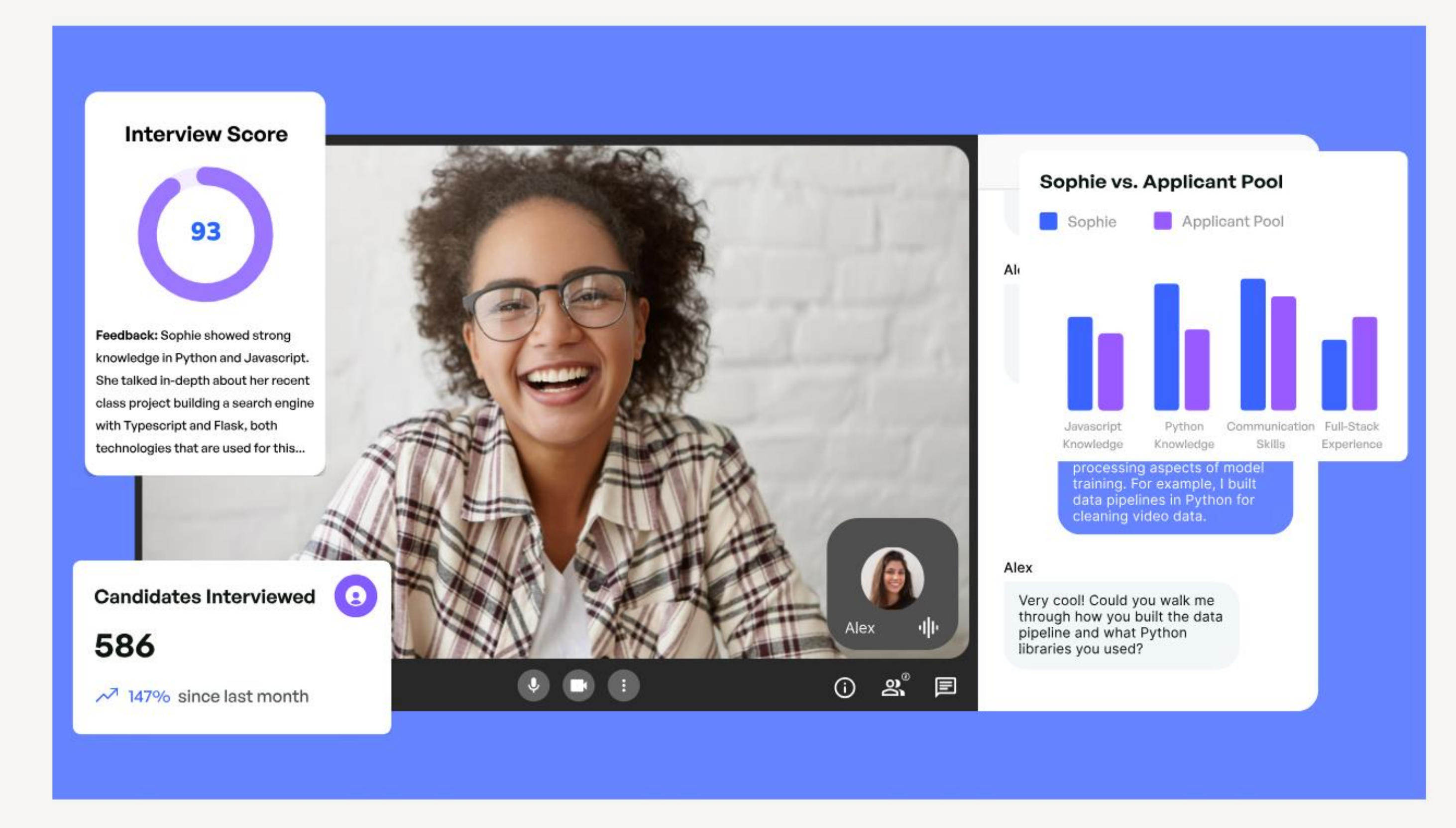
Technical Interview with AI



https://www.apriora.ai/

https://www.youtube.com/watch?v=Q7dqyoHUJ5I

How to use the AI tools responsibly?

- **As a Student/Learner:**
  - ✅ **Use AI for:** Asking for explanations, self-testing, researching, and similar activities.
  - ❌ **Don't use AI for:** Assignments, projects, tasks, or topics you're still learning.

- **In a Job or Freelancing:**
  - **Check first** if AI tool usage is permitted and which tools are allowed.
  - **Use AI to increase productivity** and improve your work.
  - **Always review** AI-generated content carefully.
  - **Don't rely on AI** for tasks you have no prior knowledge of.

Introducing Generative AI

AI terms

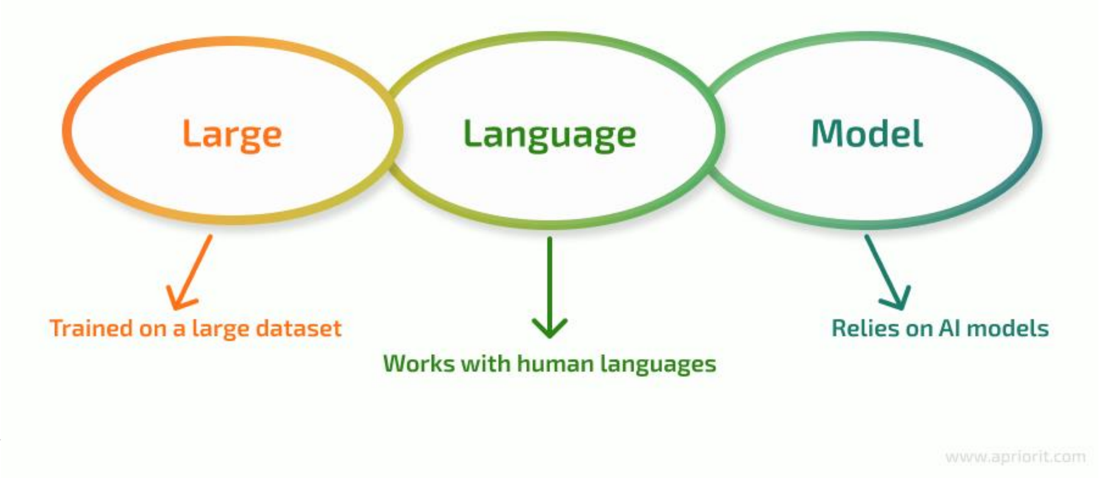
- **A Large Language Model (LLM):**
- Is an advanced AI system trained to understand and generate human-like text.
- It learns from massive amounts of text data and can answer questions, write content, and assist in conversations.

📚 Large Language Models (LLMs)

- **Definition:** A subset of Generative AI trained on massive text data to understand and generate human-like language.
- **Scope:** Focused on natural language tasks (chatting, summarizing, translating, etc.)
- **Purpose:** To process and generate text using deep learning (usually transformers).

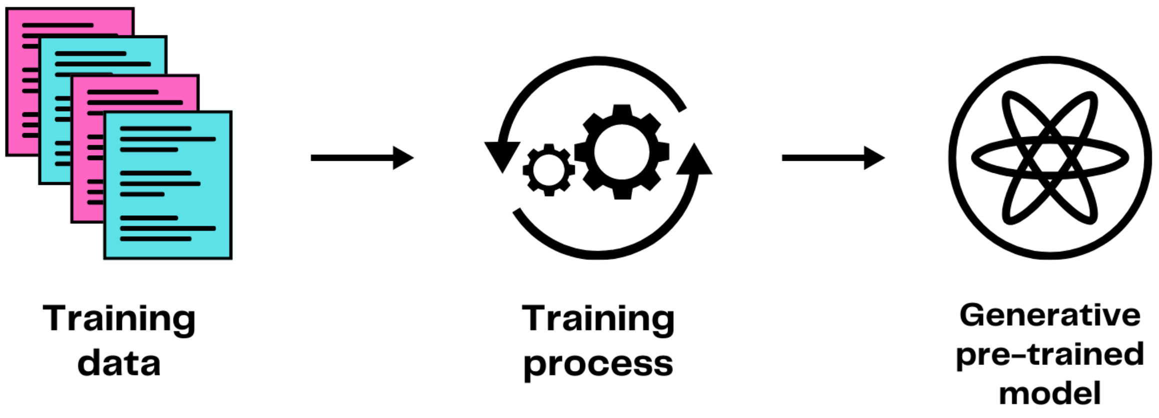
🧠 Examples:

- GPT-4 (OpenAI)
- LLaMA (Meta)
- Claude (Anthropic)



AI terms

- **AI model:** is a system trained to recognize patterns, make predictions, and automate tasks based on data. It learns from examples and improves its performance over time..
- **Multimodal models:** can process a wide variety of inputs, including text, images, and audio, as prompts and convert those prompts into various outputs, not just the source type.
- **Pre-trained Generative AI Models:** are deep learning models that have been trained on large datasets to learn general features and patterns. They serve as a foundation to enable faster development and deployment of AI. systems. GPT (Generative Pre-trained Transformer) is exampleof pre-trained generative AI models.



AI terms

- The "**Transformer**" is a deep learning architecture developed by researchers at Google, that learns context and thus meaning by tracking relationships in sequential data It's the underlying engine that makes GPT models so good at understanding and generating human language.
- The "Transformer" is the innovative architecture to allow AI models to understand the relationships between all words in a sequence simultaneously, regardless of their position. This critical feature enabled the creation of large, powerful language models like ChatGPT, allowing them to understand context deeply, learn efficiently from vast data, and generate remarkably human-like and coherent text.

AI terms

- **Reasoning Model** is a type of AI system designed to simulate human logical thinking. It processes information through a set of rules or logic to draw conclusions, solve problems, or make decisions.
- **Deep Thinking** is the ability to analyze complex problems, explore multiple perspectives, and make well-reasoned decisions.

AI terms

- **Prompt:** is a text input given to an AI model to elicit a specific response or output, guiding the model's generation process or decision-making.

- **Completion:** is the output or response generated by an AI model based on a given prompt, representing the model's attempt to complete, answer, or extend the input it received.

- **Tokens:** For text tokens Azure OpenAI processes text by breaking it down into tokens. For example: 1,000 tokens is about 750 words.

- **Prompt engineering** is the process of structuring an instruction that can be interpreted and understood by a generative AI model. A prompt is natural language text describing the task that an AI should perform.

AI terms (Cont.)

💬 Prompt

- The **input or instruction** given to a generative AI model.
- Example: *"Write a bedtime story about a robot and a cat."*

✍️ Completion

- The **AI-generated response** to a prompt.
- Example: *"Once upon a time, a robot named Zeno found a lost cat in a scrapyard..."*

🔢 Tokens

- Units of text (words or parts of words) used by the model to process input/output.
- Example:
  - "ChatGPT is smart" = 4 tokens
  - Most English words ≈ 1–2 tokens

🛠️ Prompt Engineering

- The skill of crafting clear, specific, and effective prompts to get the best results from AI.
- Example:
  - ❌ Bad: *"Tell me something."*
  - ✅ Good: *"Give me 3 pros and cons of online learning for university students."*

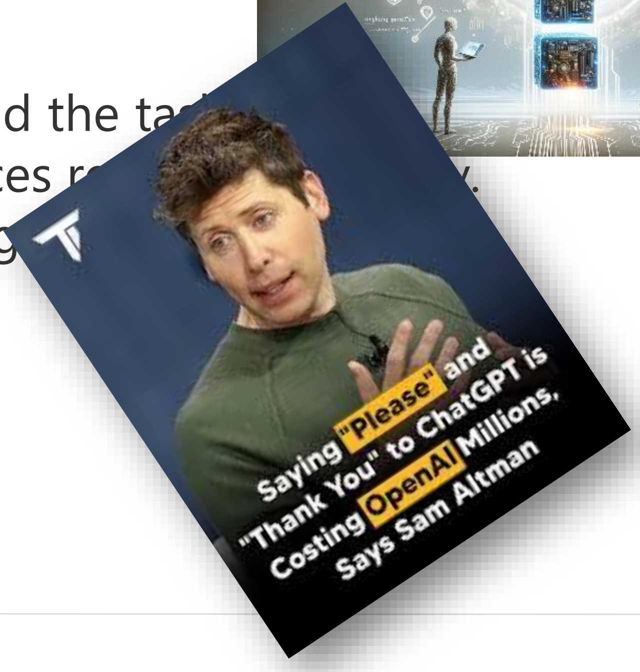
Prompt Engineering

Introducing prompt engineering

Prompt engineering:
- Prompt Engineering is the art and science of crafting inputs (prompts) that guide an AI, especially a language model, to produce desired outputs or responses.

Why It Matters?
- The quality of the prompt directly influences the AI's response.
- Good prompts lead to more accurate, relevant, and useful outputs.

Key Elements:
- **Clarity**: Clear, concise instructions help the model understand the ta
- **Context**: Providing relevant background information enhances r
- **Creativity**: Creative prompts can elicit more insightful, engag



Challenges:
- Balancing specificity and openness in prompts.
- Avoiding biases and ensuring ethical AI interactions.

Elements of a Prompt

A prompt contains any of the following elements:

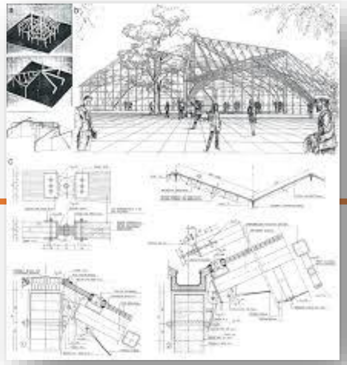
- **Instruction** - a specific task or instruction you want the model to perform
- **Context** - external information or additional context that can steer the model to better responses
- **Input Data** - the input or question that we are interested to find a response for
- **Output Indicator** - the type or format of the output.

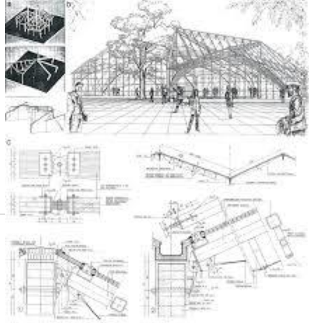
Sample Prompt:

- **Instruction**: Summarize the article.
- **Context**: The article is from a scientific journal, and it discusses the latest advancements in renewable energy technology.
- **Input Data**: [Insert link to or text of the article here]
- **Output Indicator**: Provide a concise summary in bullet points, focusing on key findings and technological innovations.

Prompt Engineering

Design and structure





Prompt Engineering’s Design and structure

1. Subject-Focused Prompts

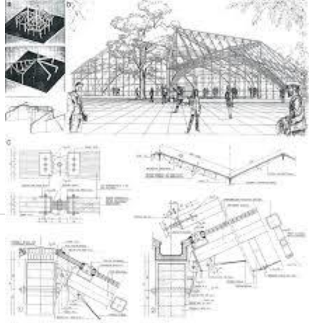
The prompt is centered on a specific topic or domain of knowledge.

The AI is expected to generate information primarily about the subject itself.

Use Case: When the user wants focused, detailed information or a summary about a specific topic.

Examples:

Explain the concept of inheritance in object-oriented programming using java



Prompt Engineering's Design and structure

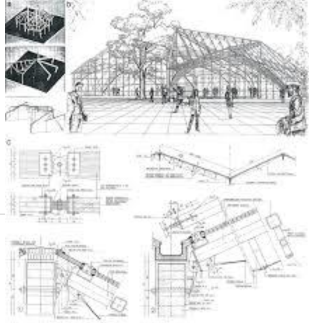
2. Problem-Focused Prompts

The prompt is framed around a problem that needs solving. It presents a challenge or question where the AI helps by providing solutions, strategies, or explanations.

Use Case: Useful for coding issues, debugging, troubleshooting, or decision-making tasks.

Examples:

How can I fix a NullPointerException error in Java?



Prompt Engineering’s Design and structure

3. Learner-Focused Prompts

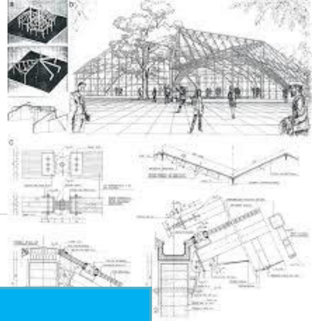
The prompt is tailored to the learner’s level, background, and needs. It adapts content delivery to match the learning style or pace of the person using the AI.

Goal: Support personalized learning, scaffolding, and interactive teaching.

Use Case: Ideal for education, tutoring, onboarding beginners, or creating step-by-step guides.

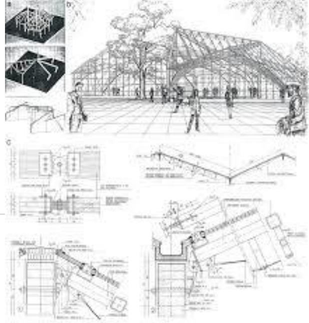
Examples:

I'm new to java programming. Can you explain what a variable is with simple examples?



Prompt Engineering's Design and structure

Feature	Subject-Focused Prompt	Problem-Focused Prompt	Learner-Focused Prompt
Primary Purpose	Provide information on a specific subject	Address and solve a specific problem	Adapt to user's learning needs
Typical Output	Definitions, explanations, summaries	Solutions, steps, troubleshooting	Simplified explanations, examples, tutoring
Target Audience	Anyone needing subject details	Problem-solvers, developers, analysts	Learners, beginners, self-learners
Prompt Tone	Neutral, academic, fact-based	Goal-oriented, directive	Supportive, adaptive, friendly
Example Prompt	**"List types of machine learning algorithms"**	**"Why is my Python code giving a syntax error?"**	**"Teach me Python like I'm 12 years old"**



Prompt Engineering’s Design and structure

Subject-focused = "Tell me *what* this is about."

Problem-focused = "Help me *solve* this."

Learner-focused = "Teach *me* based on *my* level."

Blends these styles

“I'm a beginner in Java and trying to understand how functions work. Can you first explain what a function is in simple terms, then show me a basic example, and finally help me fix null pointer exception I'm getting when I try to call my function?"

Prompt Part	Focus Type
"I'm a beginner in Python"	**Learner-focused**
"explain what a function is in simple terms"	**Subject-focused**
"show me a basic example"	**Subject-focused**
"help me fix this error I'm getting when I try to call my function"	**Problem-focused**

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Prompt Engineering

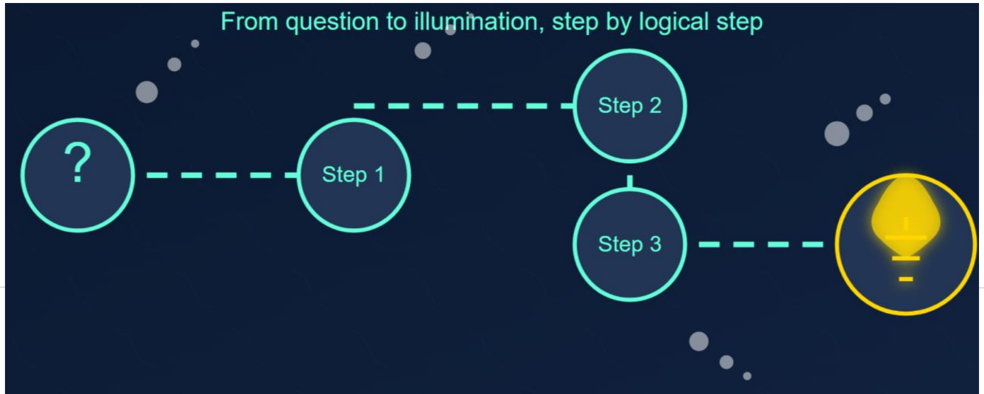
Techniques and Approaches

Prompt Engineering techniques and approaches [CoT- kot]

CoT (Chain of Thought):

- It involves structuring prompts in a way that leads the AI through a step-by-step reasoning process.
- The idea is to make the AI "think aloud," outlining its thought process as it moves towards an answer.
- Instruct the AI to show its step-by-step reasoning process before arriving at the final answer. This forces the model to "think" logically and often leads to more accurate and reliable results for complex problems.

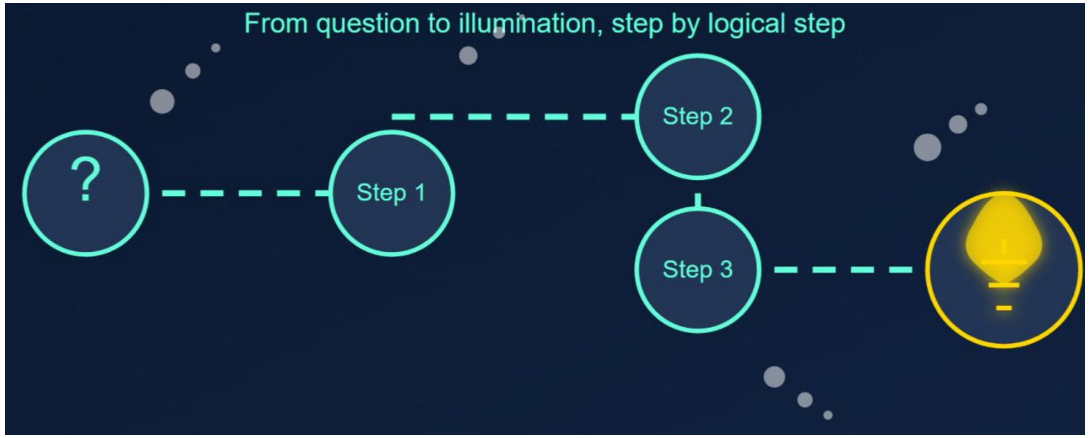
- **Approach:** Include phrases like "Think step-by-step," "Show your work," or "Reasoning process:"



Prompt Engineering techniques and approaches [CoT- kot]

CoT (Chain of Thought):

Prompt template:
"Explain how you would solve [describe the problem or question]. Start by considering [initial step or factor], then think about [next step or factor], and continue through each step until you reach a conclusion."



Prompt Engineering techniques and approaches [CoT - kot]

CoT (Chain of Thought):

Examples:

"Outline the steps you would take to increase sales of a new smartphone. Begin by identifying the target market for the smartphone. Then, think about the key features that would appeal to this market. Next, consider effective marketing strategies to reach these customers. Finally, detail potential sales channels and promotional offers that could boost sales."

CoT (Chain of Thought): (Example)

A baker makes 5 dozen cookies. She sells 45 cookies. How many cookies does she have left? Show your step-by-step reasoning.

Let's break this down step-by-step to figure out how many cookies the baker has left.

Step 1: Understand what "dozen" means.
A baker's dozen is typically 12. So, one dozen cookies equals 12 cookies.

Step 2: Calculate the total number of cookies the baker made.
The baker made 5 dozen cookies.
Total cookies = 5 dozens * 12 cookies/dozen
Total cookies = 60 cookies

Step 3: Determine how many cookies were sold.
The problem states that she sold 45 cookies.

Step 4: Calculate the number of cookies remaining.
To find out how many cookies she has left, subtract the number of cookies sold from the total number of cookies made.
Cookies left = Total cookies - Cookies sold
Cookies left = 60 cookies - 45 cookies
Cookies left = 15 cookies

Answer: The baker has 15 cookies left.

Prompt Engineering techniques and approaches [Zero-shot prompting]

Zero-shot prompting:

- It involves presenting a task to the AI model without any prior specific examples or training for that task.
- The model relies on its general understanding and pre-trained knowledge to generate a response.

When to Use Zero-shot Prompting

1. **Simple Tasks**
   → Like summarizing, translating, or generating a short description.
   Example: "Translate this sentence to Spanish."
2. **You Don't Have Examples Ready**
   → Quick tasks without time to design prompts with examples.
   Example: "Write a tweet about climate change."
3. **You Want to Test the Model's General Understanding**
   → Useful for exploring how well the AI handles new or broad tasks.
   Example: "Suggest a birthday gift for a 10-year-old who loves science."
4. **Prototyping or First Drafts**
   → When you're brainstorming or generating rough content fast.
   Example: "Create a story plot for a sci-fi movie."

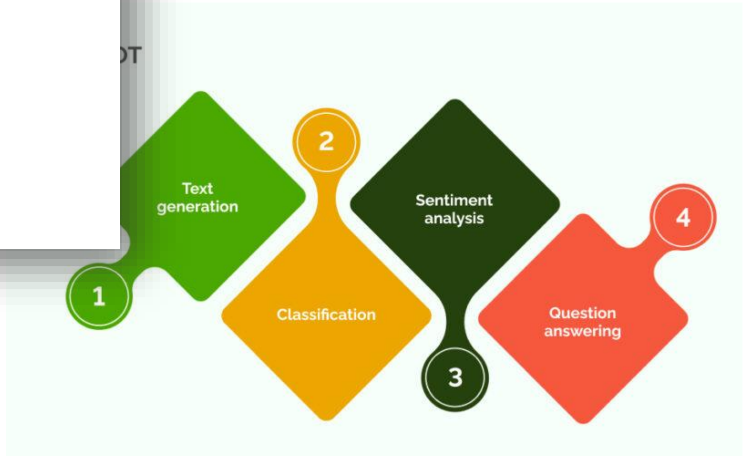
Examples:

"Assess the impact of social media on teenage mental health."

"Translate 'Hello, how are you?' into French."

"Summarize this article in 3 bullet points."

- بدون ما تشرح له إزاي يلخص
- بدون ما تدي مثال
- هو بيحاول يفهم المطلوب من أول مرة



Prompt Engineering techniques and approaches [Zero-shot prompting]

Zero-shot prompting:

- I'm providing responses without being specifically trained on the prompts.
- Zero-shot learning allows the model to generalize to tasks it hasn't seen during training by leveraging its knowledge and understanding of language and concepts.

Prompt	Response
Translate "hello" from English to French.	Bonjour
What is the capital of Australia?	Canberra
Who wrote the play "Romeo and Juliet"?	William Shakespeare
Convert 5 miles to kilometers.	5 miles is approximately 8.05 kilometers.
What is the chemical symbol for water?	H2O
Name the four primary colors.	The four primary colors are red, blue, yellow, and green.
Who is the lead singer of the band "Coldplay"?	Chris Martin
What is the currency used in Japan?	Japanese Yen
Who is the author of the "Harry Potter" series?	J.K. Rowling
Calculate the square root of 64.	The square root of 64 is 8.

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Prompt Engineering techniques and approaches

Zero-shot prompting & CoT (Chain of Thought)

Convert 10 kilograms to pounds. Let's think step by step.

Help me calculate the total cost of painting a room that is 15 feet by 16 feet with 8-foot high walls, if a gallon of paint costs $30, and each gallon covers 400 square feet? Let's think step by step.

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Prompt Engineering techniques and approaches [One-shot prompting]

One-shot prompting:

- Giving the model a single example of the task you want it to perform — usually in one prompt — to guide its response.
- But in a broader or practical context, especially when designing user-AI interactions "one-shot style" also refers to writing a single, self-contained prompt that: Includes all the context.
- Combines the question, user background, and the task.
- Doesn't require follow-up or multiple turns.

I need short, bullet-point answers to my questions.

Question: How does a home air conditioner unit work?
Answer:
- Refrigerant Circulation
- Heat Transfer
- Cooling and Dehumidification
- Thermostat Control

Question: How does an automobile air bag work?
Question: How does an artificial hear work?
Question: How does a automatic card shuffler work?
Question: How does a database work?
Question: How does a toothbrush work?
Question: How does psychology work?

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Prompt Engineering techniques and approaches [Few-shot prompting]

Few-shot prompting:

- It provides the AI model with a limited number of examples to 'learn' from before responding to a new task.
- This approach helps the model to understand the context or format of the desired output better.

Use cases

- Useful when a bit of context greatly aids the model's performance.
- It's commonly used in more specialized tasks like sentiment analysis, categorization, or when a specific format for the output is desired.

Prompt Engineering techniques and approaches [Few-shot prompting]

Few-shot prompting

Examples:

"Convert the following phrases into polite customer replies:"

Customer: "Your service was bad."
→ Reply: "We're really sorry to hear that. We'll do our best to improve!"

Customer: "I got the wrong item."
→ Reply: "Apologies for the mix-up! We'll fix this right away."

Now try this one:

Customer: "My delivery is late."

Reply: "We're very sorry for the delay. We're checking on it now and will update you shortly!"

replay Customer: "My delivery is late."

ıre, here's a professional and empathetic reply:

gent:

ıi [Customer Name], I'm really sorry your delivery is late — I understand how frustrating that can be. Let me ıeck the status right away and update you with the latest information. Thank you for your patience!"

Prompt Engineering techniques and approaches [Few-shot prompting]

Few-shot prompting

Examples:

"Generate interview questions for a [Software Developer] role with an emphasis on [backend development]. Example 1: 'Can you explain the difference between REST and SOAP APIs?' - Technical. Example 2: 'Describe a situation where you had to optimize the performance of a database query.' - Behavioral. Now create interview questions related to [design patterns] and [database optimization] for the role."

Technical – Design Patterns

1. Which design pattern would you use to implement a logging mechanism in a backend application, and why?
2. Can you explain the difference between the Singleton and Factory design patterns with examples from your past work?
3. When would you choose the Repository pattern in a backend system? What are its advantages and trade-offs?
4. Have you ever used the Strategy pattern to make your code more flexible? Walk us through the scenario.
5. How do design patterns help in achieving clean architecture in backend systems? Can you give a real-world example?

Technical – Database Optimization

1. How would you identify and resolve a slow-performing SQL query in a production environment?
2. Can you explain the impact of indexing on query performance? When might too many indexes become a problem?
3. Describe the steps you'd take to normalize a database schema and the situations where you'd prefer denormalization instead.
4. How do you handle large datasets in queries to avoid performance bottlenecks (e.g., pagination, batching)?
5. Have you worked with query execution plans? How did you use them to improve database performance?

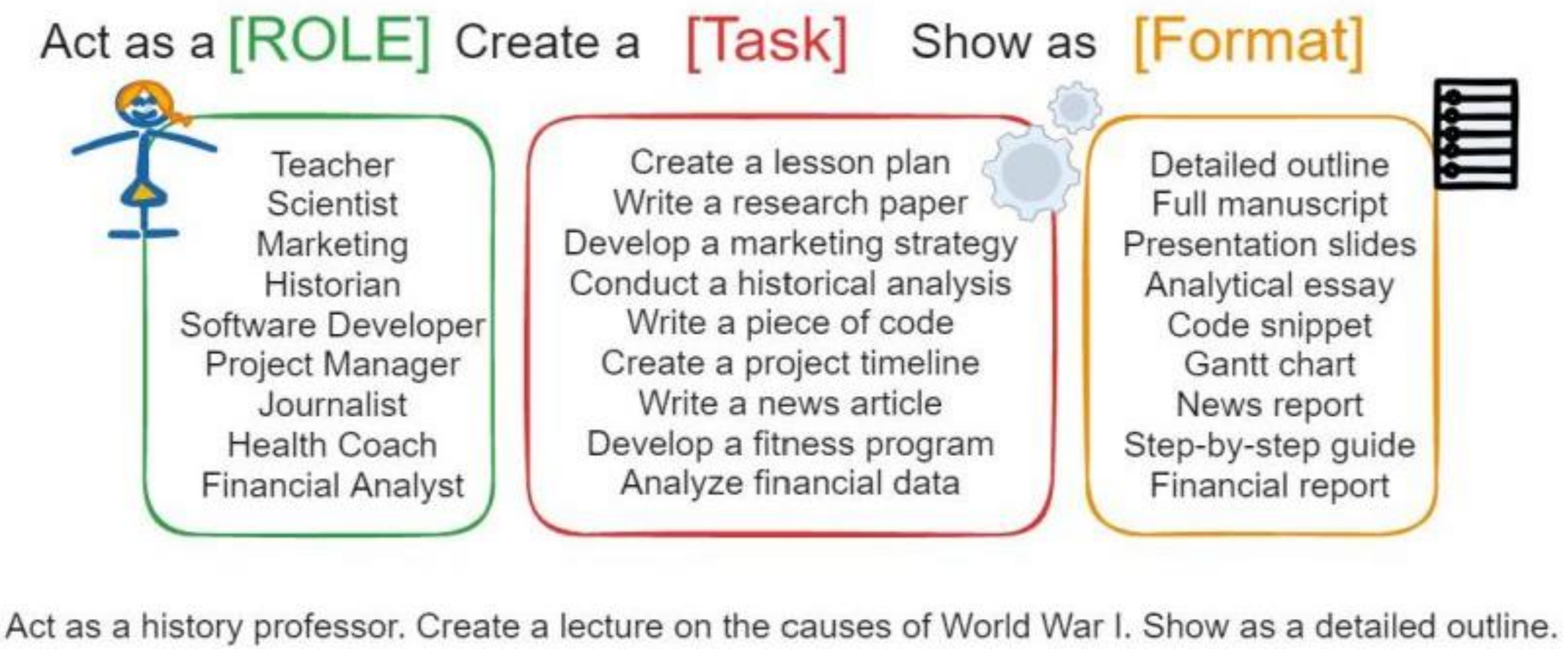
Behavioral

1. Describe a time when you had to refactor backend code to incorporate a design pattern for maintainability. What was the result?
2. Tell us about a situation where a database performance issue was affecting end users. How did you handle it and what did you learn?
3. Have you ever disagreed with a team member about using a specific design pattern? How did you resolve the disagreement?

Prompt Engineering techniques and approaches [RTF]

RTF (Role, Task, Format): one of the most know approaches, that guides the construction of prompts by focusing on three key aspects:

- **Role**: Defining the persona or capacity in which the AI model should respond.
- **Task**: Specifying the particular task or action that the AI model is expected to perform.
- **Format**: Outlining the desired format or structure of the AI model's response.



Prompt Engineering techniques and approaches [RTF]

RTF (Role, Task, Format):

Prompt Engineering techniques and approaches [RTF]

RTF (Role, Task, Format):

Examples:

“Act as a skilled software developer specializing in Java. Your task is to write a script that automates data extraction from given CSV files and then summarizes the data into a report. Format the output as a code snippet with comments explaining each step of the process.”

"As a cybersecurity expert, explain the risks of phishing attacks to a non-technical audience, providing practical advice on how to avoid them."

TF (Task, Format) only: (Example)

List the top 5 programming languages for data analysis in 2025. Present the information as a markdown table with columns for 'Language' and 'Primary Use Case'."

Please note: While I can provide an informed perspective based on current trends and the rapid evolution of technology, predicting "top" lists precisely for a future date (like 2025) involves some estimation. The landscape of programming languages is dynamic!

Here are the top 5 programming languages for data analysis as of what we can anticipate for 2025, presented in a markdown table:

Language	Primary Use Case
Python	**All-round Data Analysis & Science:** Widely used for data manipulation (Pandas), numerical computing (NumPy), statistical modeling (SciPy, StatsModels), machine learning (Scikit-learn, TensorFlow, PyTorch), visualization (Matplotlib, Seaborn), and deep learning. Its versatility and extensive libraries make it dominant.
R	**Statistical Analysis & Visualization:** A statistical programming language specifically designed for statisticians and data miners. Excellent for advanced statistical modeling, hypothesis testing, complex data visualization (ggplot2), and academic research.
SQL	**Data Extraction & Management:** Essential for querying, manipulating, and managing data stored in relational databases (e.g., PostgreSQL, MySQL, SQL Server). Crucial for preparing data for analysis and integrating with other tools.
Julia	**High-Performance Numerical & Scientific Computing:** Gaining traction for its speed, which rivals C/Fortran, while offering Python-like syntax. Ideal for computationally intensive tasks, large-scale simulations, and machine learning where performance is critical.
Scala	**Big Data Processing & Distributed Computing:** Predominantly used with Apache Spark for large-scale data processing, real-time analytics, and building robust data pipelines. Its functional programming paradigm and JVM compatibility make it a strong choice for big data engineers and data scientists working in distributed environments.

Export to Sheets

RTF (Role, Task, Format): (Example)

As a teacher, List the top 5 programming languages for data analysis in 2025. Present the information as a markdown table with columns for 'Language' and...

Alright class, let's talk about the essential tools in the data analyst's toolkit! As we move through 2025, the landscape of data analysis continues to evolve, but some languages remain steadfastly at the top due to their versatility, community support, and extensive libraries.

Here are my picks for the top 5 programming languages for data analysis in 2025, presented as a markdown table as requested:

Language	Primary Use Case
Python	**Most versatile:** Data manipulation (Pandas), numerical computing (NumPy), visualization (Matplotlib, Seaborn), machine learning (Scikit-learn, TensorFlow, PyTorch), web scraping, automation, and general-purpose programming.
R	**Statistical analysis and visualization:** Advanced statistical modeling, robust data visualization (ggplot2), academic research, and bioinformatics.
SQL (Structured Query Language)	**Data extraction and management:** Querying and manipulating data in relational databases (e.g., MySQL, PostgreSQL, SQL Server), essential for nearly all data professionals.
Julia	**High-performance scientific computing and machine learning:** Emerging choice for computationally intensive tasks, quantitative finance, and large-scale data processing due to its speed.
Scala	**Big Data processing (Apache Spark):** Ideal for large-scale, distributed data processing and real-time analytics, especially within the Apache Spark ecosystem.

Export to Sheets

A few notes on why these are top contenders:

- **Python:** Its sheer breadth of libraries and ease of learning make it the go-to for almost any data-related task, from simple analysis to complex machine learning.
- **R:** While Python has gained ground, R remains the gold standard for statisticians and researchers due to its deep statistical packages and unparalleled visualization capabilities.
- **SQL:** This isn't just a "programming language" in the traditional sense, but it's absolutely indispensable for anyone working with data. You can't analyze data you can't access!
- **Julia:** This is the rising star. Its "two-language problem" solution (combining ease of use with C-like speed) makes it increasingly attractive for performance-critical data tasks.
- **Scala:** With the continued dominance of Big Data, especially via Apache Spark, Scala's native integration and performance make it crucial for large-scale data engineering and analysis.

While other languages like Java (for enterprise Big Data), SAS (in legacy systems), and even Excel (for basic analysis) still have their places, these five offer the most robust and future-proof skill set for a data analyst in 2025. Learning at least Python and SQL will give you a significant advantage!

Prompt Engineering techniques and approaches [RISEN]

RISEN (Role, Instructions, Steps, End goal, Narrowing):

- It is a prompt engineering technique designed to break down complex or constrained tasks into actionable components.
- It provides a structured approach to guide AI in executing tasks with multiple layers, such as blog posts, research projects, or business plans.

Prompt template:

"Role: [insert the role you want AI to take.]
Instructions: [Insert the task you want AI to complete.]
Steps to complete task: [Insert numbered list of steps to follow.]
Goal: [Insert goal of the output]
Constraints: [Enter constraints].“

RISEN (Role, Instructions, Steps, End goal, Narrowing):

Example:

"Act as a [Marketing Strategist]. Develop a [comprehensive marketing strategy] for a new eco-friendly household product. Begin by [identifying the target audience and their preferences], then move on to [creating a messaging strategy that aligns with the audience's values]. Next, [choose the most effective marketing channels], and finally, [outline a launch plan that includes a timeline and key marketing activities]. Ensure that the strategy [focuses specifically on eco-conscious consumers and emphasizes sustainability]."

Prompt Engineering techniques and approaches [RISEN]

RISEN (Role, Instructions, Steps, End goal, Narrowing):

Example:

R - Role/Persona: You are a senior Python developer debugging a complex script.

I - Instructions: Explain the following Python error and suggest common causes and solutions.

S - Stimulus:Error Message:Traceback (most recent call last):
File "my_script.py", line 10, in <module>
result = data['value'] / 0
ZeroDivisionError: division by zero
Relevant Code Snippet:Python

```
data = {'name': 'test', 'value': 100}
divisor = 0 # This line is causing the issue
result = data['value'] / divisor
print(result)
```

E - Examples (Optional, but good for structure):Example 1 (Error Explanation):Input: NameError: name 'x' is not definedOutput: This error occurs when you try to use a variable or function that hasn't been defined or imported yet. Common causes include typos, forgetting to declare a variable, or not importing a required module. Solution: Check spelling, define the variable, or add an import statement.

N - Nuances/Constraints:Explain the error in simple terms, suitable for an intermediate developer.
Provide at least two common causes.
Suggest concrete solutions, potentially including code examples for the fix.
Assume the user knows basic Python syntax.
Suggest preventing such errors in the future.

Prompt Engineering Examples

Create a data set with attributes as temperature in Fahrenheit, temperature category, humidity and percentage, wind speed in miles per hour, wind direction, visibility, wind chill, precipitation, fog, rain, and snow as categorical, type in yes or no categories, month, year, and specify the data has to be created for 100 observations in a CSV format.

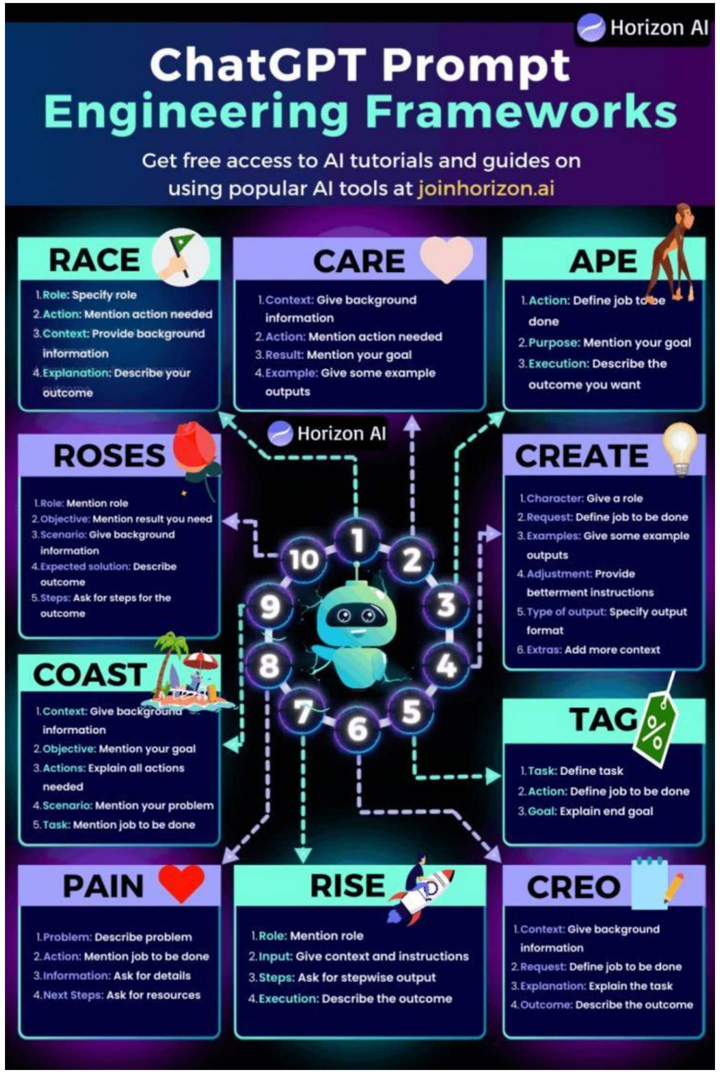
The dataset with 100 weather observations has been created. You can download it using the link below:

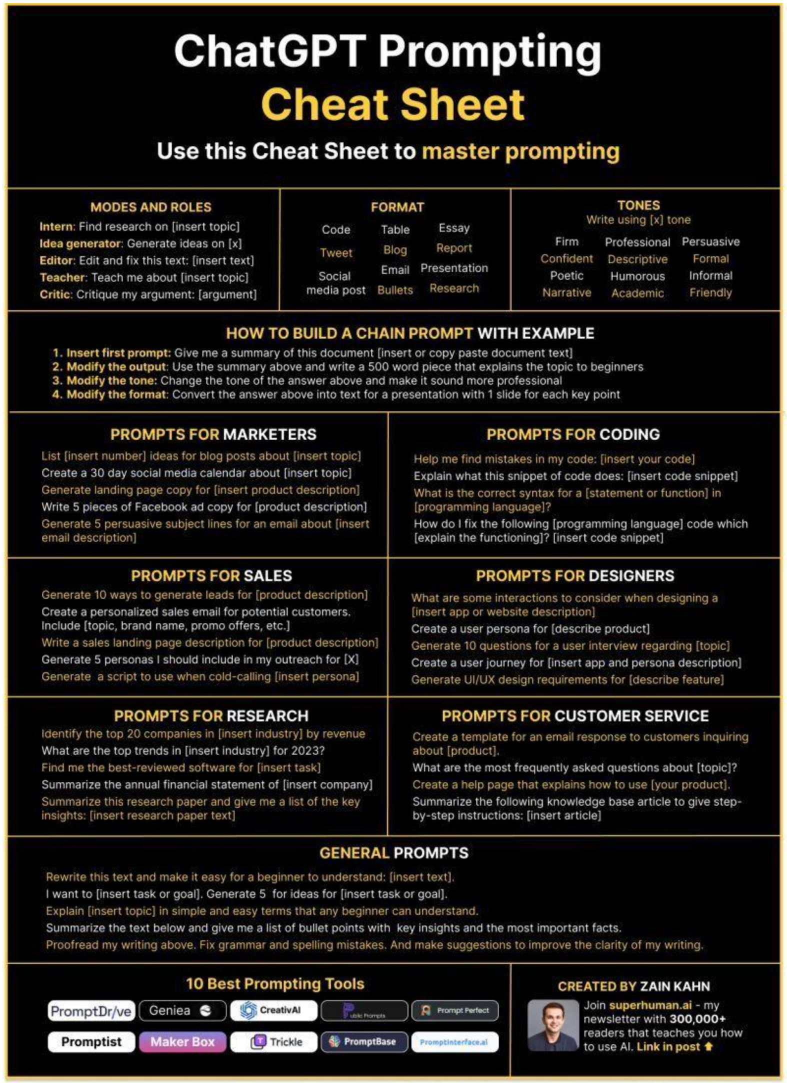
Download weather_data_100_observations.csv weather_data_100_observations.csv - Excel

	Temperature_F	Temperature_Category	Humidity_%	Wind_Speed_mph	Wind_Direction	Visibility_miles	Wind_Chill_F	Precipitation_inches	Fog	Rain	Snow	Month	Year
2	56.2	Mild	82	7.7	W	1.4	55.4	0.05	yes	no	yes	6	2025
3	96.6	Hot	71	19.8	W	1.4	89.6	0.75	no	no	yes	2	2015
4	81.2	Hot	23	24.5	SW	8.7	80.5	1.62	yes	yes	no	10	2025
5	71.9	Mild	42	16.7	E	7.3	63.7	1.97	yes	no	no	5	2025
6	40.9	Cold	34	15.9	N	5.3	33.8	0.3	no	no	no	11	2025
7	40.9	Cold	62	7.3	NE	1.9	40.1	1.19	no	no	no	10	2017
8	34.1	Cold	48	2.8	W	5.4	34.1	0.76	yes	yes	no	1	2020
9	90.6	Hot	55	26.9	SW	5.3	89.8	1.94	no	no	yes	6	2022
10	72.1	Mild	32	27	S	2.6	62.2	1.68	yes	no	no	6	2023
11	79.6	Hot	51	19	SW	4.9	75.9	1.68	yes	no	yes	4	2021
12	31.4	Cold	90	10.2	NE	4.6	31.4	0.94	yes	yes	yes	8	2018
13	97.9	Hot	78	10.5	NE	6.5	94.2	0.83	no	no	no	4	2020
14	88.3	Hot	47	21.8	SW	6.7	80.2	0.55	yes	yes	no	8	2019
15	44.9	Cold	85	26.9	N	1.4	35.4	0.11	yes	yes	yes	8	2022
16	42.7	Cold	61	26.6	SE	4.4	32.8	1.73	yes	yes	yes	5	2015
17	42.8	Cold	64	23.4	NE	6.6	35.3	1.63	no	yes	yes	4	2015
18	51.3	Mild	81	19.3	SW	5.5	47.5	2	no	yes	no	12	2025
19	66.7	Mild	76	2.5	W	8.7	65.9	1.99	no	yes	no	2	2020
20	60.2	Mild	25	4.8	SW	6.9	52.4	1.11	no	no	no	11	2024
21	50.4	Mild	47	27	E	2.5	44.8	1.54	no	yes	no	3	2016
22	72.8	Mild	47	18.2	SE	1.6	68.6	1.89	yes	no	no	3	2019
23	39.8	Cold	63	0.3	S	6.8	39.8	1.7	no	no	yes	4	2015
24	50.5	Mild	49	3	N	1.2	41.4	0.49	no	yes	no	4	2021
25	55.6	Mild	81	19.9	NW	6.3	54.5	0.9	no	no	yes	10	2015
26	61.9	Mild	94	0.2	S	9.5	57	0.26	yes	no	yes	11	2018

weather_data_100_observations

Prompt Engineering cheatsheets





General prompting Tips

1. Use the latest model
2. Put instructions at the beginning of the prompt and use ### or """ to separate the instruction and context
3. Be specific, descriptive and as detailed as possible about the desired context, outcome, length, format, style, etc
4. Reduce "fluffy" and imprecise descriptions
5. Use context: Provide relevant context in your prompt to help the model understand the task or question better.
6. Articulate the desired output format through examples (example 1, example 2).
7. Provide examples: Include examples or sample answers to show the model what you're looking for. (Few-shot)
8. Iterative approach: Don't hesitate to iterate and refine your prompts. Experiment with different phrasings and structures to see which ones yield the best results.
9. Instead of just saying what not to do, say what to do instead
10. Code Generation Specific - Use "leading words" to nudge the model toward a particular pattern

Resource and examples:

- https://www.promptingguide.ai/
- https://llmnanban.akmmusai.pro/Introductory/What-is-a-prompt/
- https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-openai-api

Lab

Build a Travel Planner AI Assistant

🎯 Objective

Design a prompt-based Generative AI assistant that helps users plan a short vacation. The assistant should provide meaningful, structured, and engaging responses using prompt engineering techniques.

📌 Task Overview

Your team will work together to:

1. Define a **user need** (e.g., a cultural trip, beach getaway, food-focused vacation, etc.)
2. Write and test three levels of prompting:
   - **Zero-shot prompt**
   - **Few-shot prompt**
   - **Refined prompt with role + format**
3. Use a GenAI tool (like ChatGPT) to test and evaluate your prompts.
4. Reflect on how prompt design impacts the quality of the output.
5. Summarize your results in a short presentation or poster.

✍️ Deliverables

- 3 tested prompts (zero-shot, few-shot, refined)
- Output samples for each prompt
- A short reflection on what worked best and why
- A visual summary (can be slide, board paper, or digital)

🧠 Examples of Use Cases (Pick one or invent your own):

- Weekend in a historical city
- Adventure trip for teens
- Cultural tour across multiple countries
- Eco-friendly vacation for nature lovers
- Honeymoon destination suggestions
- Tech conference planner for developers
- Student budget-friendly travel plan
- AI-generated tour for photographers
- Family-friendly beach holiday
- Remote work & travel (workation) experience